MarxKapital: A Computational Framework for Marxian Value Theory with Real Empirical Calibration

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Abstract

We present MarxKapital, an open-source Julia package for the mathematical modeling of Marxian value theory: capital accumulation, the tendency of the rate of profit, the transformation of labor values into prices of production, and the sectoral dynamics of expanded reproduction. Beyond the classical two-department and generalized N-department reproduction schemes, the package implements three complementary mechanisms for coupling sectoral accumulation through the technical matrix A — capital mobility (conserving aggregate constant capital exactly by construction), an aggregate realization-pressure heuristic, and an exact per-sector quantity-rationing scheme solved by fixed-point iteration — without recourse to differential-algebraic equation (DAE) solvers. Both supply-constrained mechanisms — realization pressure and quantity rationing — operate on gross output reconstructed from the model's own state and technical matrix, not a value-added-type proxy: circulating constant capital, initially omitted, is recovered from the model's own cost identity without adding new state, and investment demand is converted to output-scale units via each sector's capital-output ratio before the technical matrix is applied to it. The rationing mechanism's convergence (270 random trials, $N \in \{6, 12, 19\}$) and its effect on cross-sector dispersion (50 random economies) are checked systematically rather than on a single hand-picked example, though not yet against real per-sector accumulation data, for which no suitable time series currently exists. Computing the rationing map's Jacobian exactly by automatic differentiation reveals structure the earlier hand derivation could not: ordered by rationing severity, the Jacobian is triangular, its spectrum is real and contained in $[-1, 0]$, and -1 is always an eigenvalue once any sector is rationed — so damped iteration converges locally for every relaxation $\alpha \in (0, 1)$, undamped iteration sits exactly on a period-doubling (flip) boundary, and the observed iteration counts are predicted by the spectrum. A numerical continuation (bifurcation) analysis locates that flip at the border-collision onset of rationing. We also add a temporal single-system (TSSI) valuation layer with a fixed production lag τ, whose price map converges to simultaneist prices under frozen data in every economy tested, with no bifurcation over the exploitation range examined. Because the accumulation dynamics are homogeneous, their steady states are balanced-growth rays; we show that the value-accounting model has no interior ones, add an optional cross-dual extension in which market prices respond to excess demand, and find that its balanced growth loses stability through a supercritical Hopf bifurcation in the mobility of variable capital in all 30 random economies tested, giving way to limit cycles, tori, coexisting attractors and chaotic accumulation dynamics (positive Lyapunov exponents) that we continue numerically and confirm by direct simulation. We further present a live, reproducible empirical pipeline that calibrates the model directly from official European statistical sources (Eurostat and AMECO, via their SDMX REST APIs), replacing the synthetic or ad hoc calibration common in the computational Marxian literature, and apply it to Portugal: a 2022 sectoral national-accounts profile (19 sectors, real value added), a coherent 2020 production-structure calibration (a technical matrix reconciled exactly against Eurostat's own published output figures, paired with 2020 labor coefficients rather than mixing vintages), and an aggregate profit-rate series for 1996–2022, which ranges 7.9%–15.1% across three deflation conventions we test — a real

but modest sensitivity, not the dramatic artifact an earlier, internally inconsistent version of this calibration produced.

That earlier inconsistency is itself part of what we report. Four defects, each found by auditing rather than assuming correctness and each disclosed alongside its fix and quantified consequence: a stock-flow conflation in the model's own output formula (a capital *stock* was being added directly to annual *flows*); a depreciation term that broke an algebraic invariance the corrected rate of exploitation now satisfies exactly, verified directly rather than asserted; a technical-matrix construction that, on audit, turned out to be reading the wrong Eurostat dataflow entirely (a trade-margins satellite table, not the symmetric input-output table), understating intermediate flows by more than an order of magnitude and explaining an implausible spectral radius and maximum profit rate reported in earlier versions of this work; and, in the dynamical model itself, a supply-side flow that omitted circulating constant capital entirely and an investment-demand term that applied the technical matrix to expenditure rather than output, both corrected by reconstructing genuine gross output from the model's own state at every solver evaluation. We treat this progression as a contribution in its own right, alongside the corrected empirical numbers, on the view that a live statistical pipeline's value lies as much in surfacing and fixing exactly this kind of error as in the numbers it ultimately reports.

1 Introduction

Empirical and computational work in the Marxian tradition faces a recurring reproducibility problem: profit-rate and organic-composition estimates in the literature are frequently produced by bespoke, non-public spreadsheets or scripts, calibrated to data sources whose vintage and exact construction are difficult to reproduce [Shaikh, 2016, Duménil and Lévy, 1993]. At the same time, existing open computational implementations of Marxian reproduction schemes tend to rely on synthetic or randomly generated sectoral data for demonstration purposes, since assembling a real technical (input-output) matrix, real sectoral value-added and employment series, and real capital stock data from official sources is itself a substantial undertaking.

MarxKapital is an attempt to close both gaps at once. It is, first, a methodological contribution: a Julia implementation of the classical two-department reproduction scheme, its generalization to N departments, the labor-value and Sraffian production-price apparatus for the transformation problem, and — the paper's central technical contribution — three distinct mechanisms for coupling N-department accumulation dynamics through the technical matrix, culminating in an exact quantity-rationing scheme in the spirit of the disequilibrium-macro literature [Barro and Grossman, 1971, Bénassy, 1975], implemented without the numerical machinery a literal differential-algebraic formulation would require. This last mechanism is validated numerically and demonstrated on a synthetic economy (Section 5.4); it has not yet been tested against real per-sector accumulation data. It is, second, an empirical contribution: a live data pipeline that fetches real sectoral national accounts, input-output tables, and macroeconomic series from Eurostat and the European Commission's AMECO database via their official SDMX REST APIs, and uses them to calibrate every quantity the model needs — constant and variable capital, the rate of exploitation, the organic composition, the technical matrix itself, and (optionally) a perpetual-inventory capital-stock series — with no synthetic fallback unless explicitly requested.

We use this pipeline to calibrate the model for Portugal and report two sets of results: a detailed 19-sector profile for 2022, and an aggregate time series of the rate of exploitation, organic composition, and profit rate for 1996–2022. An earlier draft of this paper reported the latter series with a genuine defect: gross output throughout the model, $O_i = C_i + V_i + S_i$, added a capital *stock* (C_i) to two annual *flows* (V_i, S_i), and the aggregate time series used a flat 10% of the capital stock as its depreciation deduction rather than reported data. External review caught both. The stock-flow fix replaces C_i in every output-facing formula with $\delta_i C_i$, the capital *transferred* in a period; the depreciation fix replaces the flat 10% with AMECO's own consumption-of-fixed-capital series. Together they have a checkable algebraic consequence,

which we verify directly: the rate of exploitation $e^{\text{net}} = (Y - V - \text{CFC})/V$ is now exactly invariant under a common deflator applied to Y, V, and CFC together, as it must be, which it demonstrably was not under the earlier, buggy formula. With the fix in place, we re-examine the same valuation question the earlier draft raised — whether AMECO's mixed-valuation capital and flow series distort the measured trend — and find a real but much more modest effect than before: Portugal's profit rate ranges from 7.9% to 15.1% across three deflation conventions over 1996–2022 — 7.9–11.5% unadjusted, 9.3–12.7% GDP-deflated, 9.3–15.1% investment-deflated — moving mildly in either direction within each convention rather than dramatically, with the residual sensitivity now concentrated in the profit rate (through the classical capital-advanced denominator $C + V$) rather than in the rate of exploitation. We report the whole sequence — the bug, the fix, and the correctly-measured residual sensitivity — rather than only the final numbers, since the sequence is itself the more durable methodological lesson.

The remainder of the paper is organized as follows. Section 2 situates the model in the relevant theoretical literature. Section 3 presents the model itself, including the coupled N-department dynamics and the quantity-rationing mechanism (Section 3.3, with its exact Jacobian), a temporal single-system valuation layer (Section 3.4), a bifurcation analysis of the rationing iteration and the TSSI price map (Section 3.5), and an optional market-price (cross-dual) extension whose balanced growth, wage-share dependence and Hopf bifurcation to accumulation cycles are analyzed in Sections 3.6 and 5.6. Section 4 describes the empirical calibration pipeline and the Portuguese data. Section 5 reports results. Section 6 discusses the findings and the model's limitations. Section 7 concludes.

2 Related work

The theoretical foundations implemented here are classical: the reproduction schemes of Marx [1976], the tendency of the rate of profit discussed in Marx [1981], and the transformation of values into prices of production, for which we adopt the Sraffian linear-production-price formulation of Sraffa [1960]. Okishio [1961] established the conditions under which technical change and the choice of technique affect the profit rate, a result the package's technical-matrix machinery is built to be consistent with, though we do not implement an explicit choice-of-technique search. On the empirical side, Shaikh [2016] and Duménil and Lévy [1993] are the standard references for long-run profit-rate estimation from national accounts; our approach differs mainly in emphasizing a fully automated, re-runnable data pipeline over manual series construction. Closer in spirit is the strand of empirical Marxian work that estimates the profit rate and its components econometrically from official national-accounts series over long time spans [Basu and Manolakos, 2013], and the broader program of Marxian estimation from harmonized international input-output databases such as the World Input-Output Database; MarxKapital shares the latter's emphasis on a real technical matrix rather than a value-only aggregate calibration, but differs in coupling that matrix to explicit accumulation dynamics rather than using it for static decomposition alone. The monetary-expression-of-labor-time problem we flag in Section 5.3 is treated at length in Freeman and Carchedi [1996], which is also the founding statement of the temporal single-system interpretation (TSSI) whose valuation rule we implement in Section 3.4, following the formulation of Kliman and McGlone [1999]. Jacobians are computed by forward-mode automatic differentiation [Revels et al., 2016] and branches of fixed points continued numerically with BifurcationKit [Veltz, 2020]; the border-collision bifurcation we find at the onset of rationing is the piecewise-smooth phenomenon analyzed by Nusse and Yorke [1992]. The exact quantity-rationing mechanism introduced in Section 3.3 draws on the disequilibrium-macroeconomics literature on short-side rationing rules, in particular Barro and Grossman [1971] and Bénassy [1975], applied here to a Leontief-technology setting rather than the labor/goods markets those papers originally analyze. The market-price extension of Section 3.6 follows the cross-dual adjustment processes of classical competition, in which prices

respond to excess demand and capital to profit-rate differentials [Flaschel and Semmler, 1987, Flaschel et al., 1997]; the persistent accumulation cycles it produces are multisectoral relatives of the distributive growth cycle of Goodwin [1967], though here they arise from capital mobility rather than from the wage–employment interaction.

3 The model

3.1 The classical two-department scheme

The package's starting point is Marx's classical two-department scheme: Department I produces means of production, Department II produces consumption goods. Each department $d \in \{1, 2\}$ is characterized by constant capital C_d, variable capital V_d, and a rate of exploitation e_d, giving surplus value $S_d = e_d V_d$, organic composition $k_d = C_d/V_d$, and profit rate $r_d = e_d/(k_d + 1)$. Accumulation follows

$$\dot{C}_d = \sigma_{c,d} S_d - \delta_d C_d, \qquad \dot{V}_d = \sigma_{v,d} S_d, \tag{1}$$

where $\sigma_{c,d} + \sigma_{v,d} + \gamma_d = 1$ splits surplus value between reinvestment in constant capital, reinvestment in variable capital, and capitalist consumption, and δ_d is a depreciation rate. Throughout, C_d denotes constant capital *advanced* — a stock, the state variable accumulating in the equation above, on which organic composition and profit rate are conventionally computed. This is distinct from the constant capital *transferred* to output during a single period, $\delta_d C_d$ under the model's single depreciation rate. An earlier version of this paper called $W_d = \delta_d C_d + V_d + S_d$ "gross output"; that label was not accurate, since the model tracks only *fixed* constant capital C_d (a depreciating stock) and has no separate state for *circulating* constant capital z_d — raw materials, energy, purchased services, and other intermediate inputs consumed within the period without being capitalized as a stock. True gross output is $x_d = z_d + W_d$. Rather than leave z_d unrepresented, we recover it from the model's own cost structure: since A_{id} is defined as input i per unit of *gross output* of sector d (Section 3.2), a sector's circulating consumption is $z_d = \left(\sum_i A_{id}\right) x_d$, so

$$x_d = z_d + W_d = \left(\textstyle\sum_i A_{id}\right) x_d + W_d \implies x_d = \frac{W_d}{1 - \sum_i A_{id}}, \tag{2}$$

solved without introducing new state: x_d is reconstructed at every solver evaluation from W_d (itself a function of the state C_d, V_d) and the fixed matrix A, the same way Sraffa prices are recomputed at every evaluation rather than tracked as state (Section 3.2). Equation (2) is valid only if A represents each sector's *complete* circulating-input requirement — true by construction for the closed synthetic economies (`create_technical_matrix`) used in the simulations of Section 3.3 and Section 5.4, where every input a sector uses is, by definition, supplied by some sector inside the same $N \times N$ matrix. It is *not* generally true of the empirical technical matrix (Section 4.2), which we deliberately build from domestic-use flows only and which therefore, by definition, omits imported intermediate inputs; applying Equation (2) to that matrix would understate reconstructed output, because imported intermediate requirements are omitted from the domestic column sum entirely; writing $a_j^D = \sum_i A_{ij}^D$ and $a_j^M = \sum_i A_{ij}^M$ for the domestic and imported input shares, the true identity is $W_j = (1 - a_j^D - a_j^M) x_j$, so reconstructing with only A^D gives $\hat{x}_j = W_j/(1 - a_j^D) = x_j(1 - a_j^D - a_j^M)/(1 - a_j^D)$ — an understatement of $x_j - \hat{x}_j = a_j^M x_j/(1 - a_j^D)$, not simply $a_j^M x_j$; the bias depends jointly on the domestic and imported input shares, not on the imported share alone. We do not do this: Section 4's empirical calibration uses Eurostat's own published x_i directly and never invokes Equation (2), which is used exclusively for the closed synthetic economies of Section 3.3 and Section 5.4, where the completeness condition holds by construction.

The admissibility condition $\sum_i A_{id} < 1$ for every d (required for the denominator in Equation (2) to be positive) is not guaranteed by $\rho(A) < 1$, which bounds the matrix's spectral radius but not any individual column sum. We do not clamp this silently inside the per-evaluation calculation; a synthetic matrix's column sums are checked once, when the simulation is assembled, and any matrix that violates the condition is rescaled uniformly (the same rule already used to enforce $\rho(A) < 1$) until its largest column sum equals exactly 0.95 (the same margin used throughout the codebase for this purpose), with a warning logged — an explicit admissibility rule applied once to the matrix, not a numerical safeguard reapplied at every solver step. Across the 270 random synthetic matrices used in the convergence sweep of Section 5.4 ($N \in \{6, 12, 19\}$, density $\in \{0.2, 0.4, 0.6\}$), 50 (18.5%) required rescaling, by a factor averaging 0.81 (worst case 0.63) — a non-negligible fraction, reported here rather than left as an abstract possibility. We also checked whether rescaling itself measurably changes downstream behavior, rather than assuming it is neutral: the rescaled and non-rescaled subsets both converge in every trial (50/50 and 220/220), with the rescaled subset taking modestly more iterations on average (18.7 vs. 17.8) and showing a higher mean fraction of sectors rationed (1.00 vs. 0.90). Because rescaling is triggered by a matrix's pre-existing column sums, the two subsets already differ structurally by construction, and this descriptive comparison does not identify a causal effect of rescaling itself — we did not separately measure matrix density (as distinct from coefficient magnitude) between the subsets, so we do not assert which of the two drives the larger column sums that trigger rescaling. Section 3.3's coupling mechanisms use x_d, not W_d, as the supply-side quantity — resolving, rather than merely disclosing, the gap between a value-added-type flow and genuine gross output that an earlier version of this paper left as an open limitation. Intersectoral consistency (does Department I's wage and unaccumulated-surplus income suffice to buy Department II's output, and vice versa) is checked *a posteriori* by a dedicated consistency function, not imposed inside the differential system — a design choice the N-department generalization inherits and, in one respect, improves upon (Section 3.3).

3.2 Generalization to N departments and the technical matrix

For N sectors, the model introduces a technical matrix $A \in \mathbb{R}_{\geq 0}^{N \times N}$, with A_{ij} the value of good i required as input per unit of output of sector j, constructed either synthetically (with structural options for random, triangular, or block sparsity patterns, always rescaled to spectral radius below unity) or — our empirical contribution — from a real input-output table (Section 4). Given A and a labor-coefficient vector l, labor values follow the Leontief inverse, $\lambda = (I - A)^{-1} l$, and Sraffian production prices follow

$$p = w(1 + \bar{r}) \left[I - (1 + \bar{r}) A^{\top} \right]^{-1} l, \tag{3}$$

valid for any uniform profit rate $\bar{r}$ below the maximum feasible rate $R_{\max} = 1/\lambda_{\max}(A^{\top}) - 1$, where $\lambda_{\max}$ denotes the dominant eigenvalue. The baseline N-department accumulation system treats sectors as independent, exactly mirroring the two-department equations sector by sector; A is available to the value/price apparatus above but does not, in this baseline system, feed back into the accumulation dynamics themselves.

3.3 Coupling sectors through A: three mechanisms

The baseline independence of sectors is a known limitation of purely value-accounting reproduction models: physically, one sector's constant capital is another sector's output, and the technical matrix should constrain how much investment a sector can realize, not merely how its value and price magnitudes are computed. MarxKapital implements three progressively more demanding mechanisms to address this, all opt-in and mutually composable except where noted,

applied to a coupled system whose accumulation equation for constant capital becomes

$$\dot{C}_i = \kappa_i \, \sigma_{c,i} S_i - \delta_i C_i + \mu \, (r_i - \bar{r}) \, C_i, \tag{4}$$

where r_i is computed with constant capital valued at Sraffa production prices (using the current aggregate profit rate, clamped below $0.95\,R_{\max}$ to avoid the pricing singularity, with a labor-value fallback if the price system is ill-conditioned) rather than raw book value, so that A enters the dynamics through valuation as well as through the mechanisms below.

Capital mobility. The term $\mu(r_i - \bar{r})C_i$ reallocates investment toward above-average-profit sectors, a direct computational analogue of capital's tendency to flow toward higher returns. An earlier draft used $\bar{r} = \sum_i S_i / \sum_i (C_i + V_i)$, the aggregate profit-rate ratio, for this purpose. That choice does not generally satisfy $\sum_i \mu(r_i - \bar{r})C_i = 0$, since each r_i's own denominator includes V_i as well as C_i; the term could therefore create or destroy aggregate constant capital rather than purely reallocating it, a defect identified by external review. We fixed this by defining $\bar{r}$ as the C_i-weighted mean of the sectoral r_i instead, $\bar{r} = \sum_i r_i C_i / \sum_i C_i$ — the only choice for which $\sum_i \mu(r_i - \bar{r})C_i = 0$ holds by construction, which we verify directly (to floating-point precision, in the model's test suite) rather than merely asserting algebraically.

Aggregate realization pressure. Setting $\kappa_i \equiv \min(\rho_{\text{real}}, 1)$ uniformly across sectors, with $\rho_{\text{real}} = \sum_i \min(O_i, (AO)_i) / \sum_i (AO)_i$ and $O_i = x_i$ the reconstructed gross output of Equation (2) (Section 3.1) — not the value-added-type flow W_i an earlier version of this paper used here — dampens planned reinvestment economy-wide whenever aggregate intermediate demand implied by A and current outputs exceeds what those same outputs can supply. Because O_i is now genuine gross output, $(AO)_i$ is dimensionally the correct intermediate demand implied by current production, not an approximation built on an understated supply figure. This mechanism is a coarse, aggregate approximation — it does not distinguish which sectors are actually constrained — but requires no iterative solution.

Exact per-sector quantity rationing. The paper's central technical contribution replaces the aggregate heuristic with a genuine per-sector rationing scheme — "exact" in the sense of solving the specified proportional short-side rationing rule exactly via its fixed point, not in the sense of a full, empirically validated model of sectoral reproduction balances (the remaining simplifying assumption is stated below and revisited in Section 6.2). Under Leontief technology, a sector's planned reinvestment cannot substitute a shortage of one input with more of another; realized investment is bounded by the scarcest required input, and a sector's own shortfall reduces its effective demand for *all* its inputs, which in turn affects the rationing faced by its suppliers. This makes the problem simultaneous. Writing $\rho_i \in (0, 1]$ for the rationing factor faced by *suppliers* of good i and κ_j for the realized fraction of planned reinvestment for *buyer* j, the fixed point is

$$\kappa_j = \min_{i:\, A_{ij} > 0} \rho_i, \qquad D_i = \sum_j A_{ij} \, \kappa_j \, \Delta x_j, \qquad \rho_i = \min\left(1, \frac{x_i}{D_i}\right), \tag{5}$$

where $\Delta x_j = \sigma_{c,j} S_j \cdot x_j / C_j$ is the *desired expansion in gross output* implied by planned investment expenditure $\sigma_{c,j} S_j$, converted via sector j's current capital-output ratio C_j / x_j. An earlier version of this paper applied A_{ij} directly to investment expenditure $\sigma_{c,j} S_j$, implicitly treating a unit of investment spending as if it required the same input mix as a unit of physical output; that conflated a value flow with a production-scale variable, a dimensional problem external review correctly identified as more serious than a disclosed simplification. Δx_j resolves it: A_{ij} now multiplies an output-scale quantity, consistent with A's own definition (input i per unit of gross output of j), and the supply side x_i is genuine reconstructed gross output (Equation (2)),

not the value-added-type W_i. One simplification remains, disclosed rather than resolved: the *full* x_i is treated as available to meet investment demand D_i, though in reality output must also replace intermediate inputs already counted in z_i and meet consumption and other final demand first. Since no data source in the current pipeline provides sector-of-origin final-demand allocation, we do not net a consumption and replacement residual out of x_i before comparing it to D_i; the mechanism therefore represents an upper bound on investable supply, constraining the investment-goods side of reproduction specifically, not full sectoral reproduction balances (Section 6.2). The calibration in Section 4, separately, builds A from Eurostat's real product flows and uses the table's own published gross output x_i directly (Section 4.2) — Equation (2)'s reconstruction applies only to the simulated coupling mechanisms of this section, not to the empirical technical matrix itself.

We solve this by damped (under-relaxed) fixed-point iteration inside the right-hand side of the ODE at every solver evaluation, in exactly the same spirit as the Sraffa-price linear solve above, rather than exposing (ρ, κ) as differential-algebraic state to the solver. This choice is deliberate: a literal DAE formulation (mass-matrix or index reduction) or a general nonlinear-complementarity solver would add real numerical risk and a new package dependency for no benefit over the fixed point, which we verified converges reliably (a closed-form two-sector example with a known solution $\rho_1 = \sqrt{0.8}$, and no convergence failures across a systematic sweep of 270 random trials spanning $N \in \{6, 12, 19\}$ and matrix densities $\in \{0.2, 0.4, 0.6\}$, deliberately scaled to force genuine rationing rather than the trivial case, converging in a mean of 18.0 and a maximum of 20 iterations against a cap of 200 at the default relaxation $\alpha = 2/3$ — Section 5.4; the choice of α is itself derived from the Jacobian below and compared against alternatives in Section 5.5). We emphasize that undamped iteration is *not* reliable here: we verified by hand that a small two-sector example oscillates indefinitely between two states without under-relaxation, a concrete instance of the well-known failure mode of naive Picard iteration on min-based maps.

Mathematical structure of the fixed point. The convergence claim above is empirical, but three structural facts about the map in Equation (5), $F : \rho \mapsto \rho(D(\kappa(\rho)))$, can be established directly rather than left as an unexplained regularity. The natural first instinct — following the spectral-radius condition $\rho(A) < 1$ that already governs the linear Leontief/Sraffa system of Section 3.2 — is to look for an analogous Banach-contraction argument for F. This does not transfer: $\rho(A) < 1$ bounds a different, linear map, and, as we show below, it does not by itself bound F's local behavior. What can be established is more specific to F's own structure.

Existence is unconditional. D_i is clamped away from zero ($\max(D_i, \varepsilon)$ for a small $\varepsilon > 0$, avoiding division by zero when a sector's rationed suppliers deliver no input at all), and κ_j is defined as 1 when j's input column is empty (no supplier to be rationed by). With these two conventions stated explicitly, κ, D, $\rho(\cdot)$ are continuous on $[0, 1]^N$, so F is a continuous self-map of this compact convex set. Brouwer's fixed-point theorem therefore guarantees *a* fixed point exists for every admissible (A, σ_c, S, x), independent of whether any particular iterative scheme converges to it. Existence does not imply uniqueness: because F is antitone (below), the model could in principle admit multiple fixed points or a genuine two-cycle rather than converging under damping. We do not have a uniqueness proof; what we have is a direct empirical check, run against a real 19-sector matrix under a synthetic demand shock tight enough to bind every sector ($\kappa_i < 1$ for all i), varying the initial ρ (50 uniform draws on $(0.1, 1]^N$), the sector ordering (20 permutations, unpermuted after solving), and the convergence tolerance (10^{-4} to 10^{-12}): the damped iteration reached the same fixed point to numerical precision ($< 10^{-6}$) in every case. This is evidence of practical uniqueness in the regime we test, not a proof of uniqueness in general.

F is antitone (order-reversing) in ρ. $\kappa_j(\rho) = \min_{i:A_{ij}>0} \rho_i$ is isotone (a minimum is monotone in each argument); $D_i(\kappa) = \sum_j A_{ij}\kappa_j \Delta x_j$ is isotone (a nonnegative linear combination);

and $D \mapsto \min(1, x_i/D_i)$ is antitone (decreasing in D_i). Composing two isotone maps with one antitone map yields an antitone map. This is the structural reason undamped Picard iteration oscillates rather than diverging or slowly converging: iterating an antitone map is a setting in which two-cycles, rather than convergence, are a characteristic possible failure mode, because $F \circ F$ — not F itself — is isotone. Our two-sector example instantiates this known qualitative behavior of order-reversing maps; it is not evidence that every antitone map in this class oscillates, only that oscillation is a real and structurally-explained possibility rather than an unexplained curiosity.

Damping relocates a real eigenvalue, exactly. At an interior fixed point ρ^* (no ρ_i pinned at a boundary), differentiating Equation (5) gives Jacobian entries

$$\frac{\partial F_i}{\partial \rho_k} = -\frac{x_i}{D_i^2} \sum_{j:\, i^*(j)=k} A_{ij}\, \Delta x_j \;\le\; 0, \tag{6}$$

where $i^*(j)$ denotes a supplier attaining the minimum defining κ_j — confirming antitonicity locally, in agreement with the global argument above. For the damped step $\rho \mapsto (1-\alpha)\rho + \alpha F(\rho)$, a real Jacobian eigenvalue λ of F at ρ^* maps to $\mu(\alpha) = (1-\alpha) + \alpha\lambda$, and $\alpha^\star = 1/(1-\lambda)$ sets $\mu(\alpha^\star) = 0$: damping at $\alpha^\star$ exactly annihilates the local linear error mode associated with λ. This is a statement about one eigenvalue, not a claim that the full damped iteration converges superlinearly in general — that would additionally require every other eigenvalue of the local Jacobian to vanish or to already lie well inside the unit disc at $\alpha^\star$. An earlier version of this paper left that condition unchecked; Proposition 1 below settles it for every admissible instance. This is not hypothetical: for the closed-form two-sector example above, the Jacobian collapses to a single active eigenvalue $\lambda = -1$ exactly (verified both algebraically from the formula above and numerically), for which $\alpha^\star = 1/(1-(-1)) = 0.5$ — the relaxation value earlier versions of the package used by default, chosen before this analysis rather than tuned after the fact to make this example converge. The unrationed sector, however, contributes a second eigenvalue $\lambda = 0$, whose error mode decays at $|1-\alpha|$; the α that minimizes the spectral radius over *both* modes is therefore $2/3$ (radius $1/3$), not 0.5 (radius 0.5) — annihilating the dominant mode and minimizing the overall contraction rate are different criteria. The package now uses $\alpha = 2/3$ by default, for the reasons given after Proposition 1.

We also checked directly whether a global, $\rho(A)$-style Lipschitz bound exists for F, rather than assuming one does not. Write $L_i = \sum_j A_{ij}\Delta x_j$ for the full (unrationed) demand on good i and $r_i = x_i/L_i$. A global Lipschitz constant for F on the box $[\rho_{\min}, 1]^N$ is $K = \max_i x_i/(\rho_{\min}^2 L_i) = \max_i r_i/\rho_{\min}^2$; requiring F to map the box into itself forces $\rho_{\min} \le \min_i r_i$, and at the tightest admissible choice $\rho_{\min} = \min_i r_i$ the bound gives $K \ge 1/\min_i r_i > 1$ whenever any sector is actually rationed ($r_i < 1$ for some i) — i.e. this style of global bound is provably vacuous in exactly the regime the mechanism is designed for. We report this as a genuine negative result rather than omit it: no clean global contraction constant in the spirit of $\rho(A) < 1$ exists for this map by this route, which is why we rely on the local analysis above, together with the empirical convergence sweep reported earlier in this section (270 trials, no failures — Section 5.4), rather than a single global theorem.

The exact Jacobian is triangular. The hand derivation above covers the two-sector case; computing $J = \partial F/\partial\rho$ exactly by forward-mode automatic differentiation for general N (`rationing_jacobian`) exposed a structure that holds in general and can be proved directly.

Proposition 1. *Let ρ^* be a fixed point of F at which every κ_j has a unique minimizing supplier $i^*(j)$ and no x_i/D_i equals 1 exactly (the generic case). Order sectors by increasing ρ_i^*. Then $J = \partial F/\partial\rho$ at ρ^* is lower triangular, and its eigenvalues are $\lambda_k = -\theta_k$, where*

$$\theta_k = \frac{1}{D_k} \sum_{j:\, i^*(j)=k} A_{kj}\, \kappa_j\, \Delta x_j \;\in\; [0, 1] \tag{7}$$

is the share of sector k's effective demand that comes from buyers whose investment k itself bottlenecks ($\theta_k = 0$ for unrationed sectors). If any sector is rationed, the most rationed sector $k_0 = \arg\min_i \rho_i^*$ has $\theta_{k_0} = 1$, so $\lambda = -1$ is an eigenvalue, and $\rho_{k_0}^* = \sqrt{x_{k_0}/L_{k_0}}$ in closed form, where $L_i = \sum_j A_{ij}\Delta x_j$ is unrationed demand.

Proof. By the Jacobian formula above, $J_{ik} \neq 0$ only if $k = i^*(j)$ for some buyer j with $A_{ij} > 0$; since i is then itself one of j's suppliers and k attains the minimum over them, $\rho_k^* \leq \rho_i^*$. Ordering by ρ^* makes J lower triangular, so its eigenvalues are its diagonal entries; at the fixed point $x_k/D_k = \rho_k^* = \kappa_j$ for every j with $i^*(j) = k$, which turns J_{kk} into $-\theta_k$, and $0 \leq \theta_k \leq 1$ because the sum runs over a subset of the terms of D_k. For k_0, every buyer j that k_0 supplies has $\kappa_j = \rho_{k_0}$, so $D_{k_0} = \rho_{k_0} L_{k_0}$ and $F_{k_0} = x_{k_0}/(\rho_{k_0} L_{k_0})$ depends on ρ_{k_0} alone; hence $\theta_{k_0} = 1$, and solving $\rho = x_{k_0}/(\rho L_{k_0})$ gives the closed form, which generalizes the two-sector $\rho_1 = \sqrt{0.8}$. □

Three consequences follow. First, the damped multipliers $1 - \alpha(1 - \lambda_k) = 1 - \alpha(1 + \theta_k)$ all lie in $(-1, 1)$ for every $\alpha \in (0, 1)$: damped iteration is *locally* convergent at every generic fixed point, for every admissible (A, σ, S, x), not only in the cases swept. Second, at $\alpha = 1$ the multiplier of k_0 is exactly -1: undamped Picard iteration sits on a flip (period-doubling) boundary whenever rationing binds, which is the mechanism behind the two-cycle observed by hand, now shown to be generic rather than an artifact of one example; equivalently, the local stability boundary in α is $\alpha_{\max} = 2/(1 + \max_k \theta_k) = 1$ exactly. Third, the spectral radius of the damped step is $\max_k |1 - \alpha(1 + \theta_k)|$, and since $\theta_k \in [0, 1]$ it is bounded by the two extreme modes, $\max(|1 - \alpha|, |1 - 2\alpha|)$. Once rationing binds, $\theta = 1$ is always present (sector k_0), and $\theta = 0$ is present whenever some sector is unrationed or bottlenecks none of its buyers — in every rationed case we tested (Section 5.5). The bound is then attained, and minimized at $\alpha = 2/3$, where it equals $1/3$: the local error shrinks by a factor of three per step, against a factor of two at the previous default $\alpha = 0.5$. We adopt $\alpha = 2/3$ as the package default on this basis and check both the prediction and global behavior from far-off initial conditions in Section 5.5. What the proposition does *not* give is uniqueness of the fixed point or global convergence from arbitrary initial ρ; both remain open (Section 6.2).

None of the three mechanisms imposes exact market clearing at every instant for every sector as a hard constraint solved simultaneously with the differential state — that would require the DAE formulation we deliberately avoid. A dedicated diagnostic function generalizes the two-department *a posteriori* check to N sectors, reconstructing x_i from the state via Equation (2) and reporting the per-sector *intermediate-use residual* $x_i - (Ax)_i$ and its aggregate ratio at a given point in the trajectory. We do not call this a market-clearing gap or a disequilibrium measure: in standard input-output accounting, gross output minus modeled intermediate demand is ordinarily positive and expected — it is output available for final use, exports, inventories, and every other demand category this diagnostic does not model (Section 6.2) — so a positive residual is the normal case, not evidence of anything approaching balance. The residual is informative as a relative, within-model comparison (larger under one parameterization than another, changing over the trajectory) and now uses a genuine gross-output x_i rather than the value-added-type W_i an earlier version of this diagnostic used, but it should not be read as a sectoral equilibrium or reproduction-balance measure on its own.

3.4 Temporal single-system valuation with a fixed production lag

The Sraffian prices of Section 3.2 are *simultaneist*: inputs are valued at the same prices as the outputs they help produce. The temporal single-system interpretation [Freeman and Carchedi, 1996, Kliman and McGlone, 1999] instead values the constant capital used up in a period at the prices at which it was actually purchased. We implement this as a valuation layer on top of simulated trajectories (`src/tssi.jl`); the dynamics of Sections 3.1–3.3 are unchanged. Inputs used at t were bought at $t - \tau$, with the production lag τ held fixed. Writing quantities per unit

of reconstructed gross output x_j (Equation (2)), with $d_j = \delta_j C_j/x_j$, $v_j = V_j/x_j$, $s_j = S_j/x_j$, the temporal valuation is

$$c_j(t) = \big(A^\top p(t-\tau)\big)_j + d_j + v_j, \qquad r^{\mathrm{T}}(t) = \frac{\sum_j S_j}{\sum_j x_j c_j(t)}, \qquad p(t) = \big(1 + r^{\mathrm{T}}(t)\big)\, c(t), \tag{8}$$

with unit values $\lambda_j(t) = (A^\top p(t-\tau))_j + d_j + v_j + s_j$. Both TSSI aggregate equalities hold by construction: total price equals total value, and total profit equals total surplus value. We verify them numerically along every trajectory (maximum relative error 1.1×10^{-15}). Equation (8) is a delay-difference equation in continuous time; on a grid of step h with $\tau = mh$ it decouples into m interleaved copies of the map $G : p(t-\tau) \mapsto p(t)$.

The simultaneist counterpart uses current prices for inputs: $p = (1+r)(A^\top p + d + v)$ with $r = \sum S / \sum x(A^\top p + d + v)$. For given r, $p(r) = (1+r)[I - (1+r)A^\top]^{-1}(d+v)$, and $r \sum_j x_j p_j(r)/(1+r) - \sum S$ is strictly increasing on $[0, R_{\max})$, negative at 0 and unbounded as $r \to R_{\max}$, so the simultaneist solution (p^*, r^*) exists and is unique. It is exactly the fixed point of G under frozen data. Whether TSSI prices converge to it is therefore a question about the spectrum of

$$\left.\frac{\partial G}{\partial p}\right|_{p^*} = \left[(1+r^*)I - r^* \frac{c\, x^\top}{x^\top c}\right] A^\top, \qquad c = A^\top p^* + d + v, \tag{9}$$

which we compute both in closed form and by automatic differentiation (agreeing to 10^{-10} relative). Because the lag is a single fixed τ, stability does not depend on τ; τ only sets the time scale, with continuous-time convergence rate $\log \varrho(\partial G/\partial p)/\tau$, where ϱ is the spectral radius. Away from frozen data — along an accumulating trajectory — the two valuations differ because TSSI carries the history of input prices that simultaneism discards.

3.5 Bifurcation analysis

The accumulation ODEs grow without bound, so they have no equilibria in levels to continue; their steady states are balanced-growth rays, taken up in Section 3.6. The other objects with genuine fixed points are the two maps above: the damped rationing step $\rho \mapsto (1-\alpha)\rho + \alpha F(\rho; s)$, where s scales planned demand ($\Delta x \to s\,\Delta x$), and the TSSI price map G with fixed τ, where ϑ scales surplus value ($S \to \vartheta S$). For each, we continue the branch of fixed points with pseudo-arclength continuation (BifurcationKit, Jacobians by automatic differentiation) and classify local bifurcations from the multipliers μ of the map's Jacobian along the branch: a real μ crossing $+1$ is a fold, a real μ crossing -1 a flip (period doubling), and a complex pair crossing $|\mu| = 1$ a Neimark–Sacker bifurcation. BifurcationKit's own detectors look for eigenvalues crossing the imaginary axis, the criterion for ODEs, and would miss flips of maps; the classification is therefore done on the multipliers directly (`continue_fixed_point`).

For the rationing step, Proposition 1 predicts the diagram. Below the onset $s_c = \min_i x_i/L_i$ no sector is rationed, $F \equiv 1$ locally, $J = 0$, and the step is stable for every $\alpha \in (0, 2)$. At s_c the most-demanded sector becomes rationed and $\theta_{k_0} = 1$ appears discontinuously: the stability boundary drops from $\alpha_{\max} = 2$ to $\alpha_{\max} = 1$ in a single jump. Because F is only piecewise smooth (through the min operators), this is a border-collision bifurcation [Nusse and Yorke, 1992], not a smooth one. For any $\alpha \in (1, 2)$ it is a border-collision flip: the fixed point loses stability through a multiplier jumping past -1 exactly at s_c. For $\alpha < 1$ there is no bifurcation anywhere along the branch.

3.6 Market prices, balanced growth and accumulation cycles

Balanced growth as a fixed point. The right-hand side f of the accumulation system is homogeneous of degree one in the state $X = (C, V)$: doubling every capital doubles every flow. Trajectories therefore never settle in levels, and the economically meaningful steady states are

balanced-growth rays, $f(X^\star) = g^\star X^\star$, along which every capital grows at the common rate $g^\star$. Normalizing to the simplex, $y = X/\mathbf{1}^\top X$, gives

$$\dot{y} = f(y) - \big(\mathbf{1}^\top f(y)\big)\, y, \tag{10}$$

whose fixed points are exactly the balanced-growth rays and whose Jacobian, written in $2N-1$ free coordinates so that the simplex constraint is not a spurious zero eigenvalue, governs the stability of the economy's *proportions* (`src/growth.jl`).

No interior balanced growth without market prices. In the coupled system of Section 3.3 a sector's profit rate depends on its own organic composition k_i and on aggregates (the uniform price-of-production rate, $\bar{r}$, the growth rate), but not on its size relative to other sectors. A balanced-growth ray then requires, for each sector, the two equations $\dot{C}_i/C_i = g^\star$ and $\dot{V}_i/V_i = g^\star$ in one sector-specific unknown k_i plus a few aggregates: $2N$ equations in $N + O(1)$ unknowns, generically without solution once N exceeds a handful. Numerically, the normalized flow converges to a boundary of the simplex — capital concentrates in one sector — with every combination of mechanisms we tried, including a mobility term for variable capital $\mu_v(r_i - \bar{r}_V)V_i$ ($\bar{r}_V$ the V-weighted mean of the r_i, so that the term sums to zero) added by analogy with Equation (4).

Cross-dual dynamics. What is missing is a channel through which a sector that grows beyond the demand for its product earns less. We add one, opt-in, in the cross-dual tradition of Flaschel and Semmler [1987] and Flaschel et al. [1997]: market prices P_i respond to sectoral excess demand, and profits — hence capital mobility — are computed at those prices (`src/cross_dual.jl`). With $x_j = W_j/(1 - \sum_i A_{ij})$ the reconstructed gross output of Equation (2),

$$\begin{aligned}
\Pi_i &= \big(P_i - (A^\top P)_i\big)x_i - V_i - \delta_i C_i, & r_i &= \frac{\Pi_i}{C_i + V_i},\\
\dot{C}_i &= \sigma_{c,i}\Pi_i - \delta_i C_i + \mu\,(r_i - \bar{r})\,C_i, & \dot{V}_i &= \sigma_{v,i}\Pi_i + \mu_v\,(r_i - \bar{r}_V)\,V_i,\\
\dot{q}_i &= \beta\,\frac{d_i - x_i}{x_i}, & P_i &\propto e^{q_i}.
\end{aligned} \tag{11}$$

Prices are normalized so that total profit equals total surplus value, $\sum_i \Pi_i = \sum_i S_i$; with $P = \mathbf{1}$ the model reduces to value accounting, $\Pi = S$. Demand d is intermediate demand Ax plus investment and consumption expenditure deflated by market prices: gross investment $\sigma_{c,i}\Pi_i + \delta_i C_i + \mu(r_i - \bar{r})C_i$ buys capital goods in the composition of column i of A, and wages and capitalist consumption buy a fixed consumption basket (by default the initial net-output shares, an assumption: the pipeline has no final demand by sector of origin). Walras' law, $\sum_i P_i d_i = \sum_i P_i x_i$, then holds identically (maximum relative error 5.4×10^{-16} over random states), and the system remains homogeneous of degree one in (C, V) and invariant to a common shift of the log-prices q, so the normalized system (10) has dimension $3N - 2$. A sector that outgrows its demand sees its price, profit rate and capital fall; the resulting balanced-growth rays are interior.

Bifurcations of the accumulation ODE. The fixed points of the normalized cross-dual system are continued in a parameter — μ, μ_v, a uniform wage share ω ($e_i = (1-\omega)/\omega$), or a proportional scaling of the sectoral rates of exploitation — with pseudo-arclength continuation and Jacobians by automatic differentiation (`accumulation_bifurcation`). Unlike the maps of Section 3.5, this is a genuine ODE, and BifurcationKit's detection of eigenvalues crossing the imaginary axis applies directly. At each Hopf point we compute the normal form

$\dot{z} = z(i\omega_H + a\,\delta\theta + b|z|^2)$: $\mathrm{Re}\,a$ is the transversal crossing speed and the first Lyapunov coefficient $\ell_1 = \mathrm{Re}\,b$ decides whether the bifurcating cycle is stable ($\ell_1 < 0$, supercritical) or unstable [Kuznetsov, 2004]. From a supercritical Hopf we continue the branch of limit cycles by orthogonal collocation, with stability from the Floquet multipliers and detection of their bifurcations: a fold of cycles, period doubling, and a Neimark–Sacker bifurcation to an invariant torus. The branch is seeded from the normal-form predictor at a small distance from the Hopf point; BifurcationKit's automatic branch switching fails on this system. Every cycle is cross-checked by direct simulation of (10) with a Poincaré section (`simulated_cycle`): the spread of section crossings separates a limit cycle from quasi-periodic or irregular motion, and the return times give the period. Irregular motion is classified by its leading Lyapunov exponents (`lyapunov_exponents`), computed with the method of Benettin et al. [1980]: the variational equations, with the exact Jacobian, are integrated alongside the trajectory and re-orthonormalized by QR decomposition every time unit. Exponent signatures distinguish a limit cycle $(0, -, -)$, a torus $(0, 0, -)$, chaos $(+, 0, \dots)$ and hyperchaos (two positive exponents). Coexisting attractors are detected by sweeping the parameter up and then down with direct simulation, each step starting from the previous step's final state (`simulated_hysteresis`).

4 Empirical calibration

4.1 Data sources

MarxKapital fetches five families of real sectoral and macroeconomic data [Eurostat, 2026b,a, European Commission, Directorate-General for Economic and Financial Affairs, 2026], each via an official SDMX REST API rather than a static file:

- **Sectoral value added, compensation, and consumption of fixed capital** from Eurostat's national accounts by industry (`NAMA_10_A64`), at the 19-section NACE Rev. 2 level.
- **Sectoral employment and gross fixed capital formation** from the companion Eurostat datasets `NAMA_10_A64_E` and `NAMA_10_A64_P5`.
- **Sectoral net capital stock** (the default series used in Section 4.3) from `NAMA_10_NFA_ST`, item `N11N`.
- **A real technical matrix** from Eurostat's symmetric input-output table at basic prices, product by product (`NAIO_10_CP1700`), domestic-use flows only, aggregated to Leontief technical coefficients reconciled directly against the table's own published gross-output row (Section 4.2).
- **Aggregate macroeconomic series** (GDP, employment, compensation, capital stock, and eighteen further variables) from the European Commission's AMECO database.

All four sources were, until recently, distributed as static bulk files with heuristic, undocumented formats; both institutions have since migrated to official SDMX REST APIs, which MarxKapital uses directly, with the legacy static-file parsers retained only as a manual fallback.

4.2 Constructing the technical matrix: a second bug, found by auditing rather than assuming correctness

An earlier version of this pipeline, and of this paper, built the technical matrix from `NAIO_10_CP1620`. External review asked us to audit this construction — quantify the effect of a margin-clipping step, reconcile row/column totals against published data, document how product and industry

classifications are matched, and resolve whether imports are treated as domestic supply. Carrying out that audit (`paper/matrix_audit.jl`) surfaced something more fundamental than any of those individual questions: `NAIO_10_CP1620` is not the symmetric input-output table. Its Eurostat title is "Table of trade and transport margins" — a small satellite table used to decompose the use table at purchasers' prices into its components (basic-price flows plus margins plus taxes less subsidies), not the symmetric table itself. Building A from it produced flows understated by more than an order of magnitude: for Portugal, 2020 (the same reference year as the corrected matrix below, for a like-for-like comparison), total intermediate consumption implied by `NAIO_10_CP1620` is ≈€8.8 bn, against a true economy-wide figure of ≈€177.8 bn (published output less published value added, summed across all NACE sections). Every negative cell we had previously attributed to a margin-reallocation accounting convention was simply the mechanics of that satellite table, not a property of the real input-output structure at all. This is the direct explanation for the implausibly low spectral radius ($\rho(A) \approx 0.13$) and implausibly high $R_{\max}$ ($\approx 691\%$) reported in earlier versions of this paper, which the review correctly flagged as suspicious without yet knowing why.

The real table is `NAIO_10_CP1700`, "Symmetric input-output table at basic prices (product by product)." Rebuilding A from it, for Portugal, gives zero negative cells (not 272–317 clipped, as before) and reconciles *exactly* against the table's own published gross-output row (item `P1`, available for all 19 sections), which we now use directly as the output denominator rather than approximating it from value added and intermediate-flow sums. The corrected matrix has spectral radius $\rho(A) \approx 0.52$ and implied $R_{\max} \approx 91\%$ (Section 5.3) — both economically plausible, unlike their predecessors.

Two further questions the review raised are resolved directly rather than left as caveats. *Imports:* `NAIO_10_CP1700` publishes domestic-use (`DOM`), imports-use (`IMP`), and combined (`TOTAL`) variants — confirmed by querying the API with the `stk_flow` dimension unconstrained, something `NAIO_10_CP1620` did not offer (it published `TOTAL` only). We use `DOM`: A is meant to represent domestic inter-sectoral production dependence, and an imported input is not supplied by any domestic sector, so including it in A would assert a domestic supply relationship that does not exist. *Classification:* the symmetric table is genuinely product-by-product, using the same CPA product code on both the row (`prd_ava`) and column (`prd_use`) dimensions at the resolution we use — there is no CPA-to-NACE concordance to document for this step, since no NACE-classified dimension is involved in constructing A itself; NACE enters only through the separately-sourced sectoral value-added weights (`NAMA_10_A64`), a different and already-documented join.

One limitation the audit surfaced and we chose to resolve by scoping, rather than by leaving a mismatch in place: `NAIO_10_CP1700` is a benchmark table, published for specific reference years (2010, 2013, 2015, 2017, 2020 for Portugal, not annually), not a series like the rest of the pipeline. Requesting a matrix for 2022 silently falls back to the nearest available reference year — 2020 for Portugal, logged explicitly as a warning at calibration time — which would mix a 2020-vintage matrix with 2022 labor coefficients if left uncorrected. We avoid that mismatch directly: Figures 3 and 4 (the heatmap and the transformation-problem comparison) use a fully coherent 2020 calibration, matrix *and* labor coefficients both from 2020, not a mixed vintage. Table 1's sectoral e, k, r, and GVA figures do not depend on A at all and remain genuinely 2022 (Section 5.1), as does the aggregate 1996–2022 AMECO series (Section 5.2); neither needed to change. The paper's results therefore span two internally-coherent reference points — a 2020 production-structure snapshot (Figures 3–4) and a 2022 sectoral/aggregate snapshot (Table 1, Figure 1) — rather than one section silently combining both.

4.3 Depreciation and capital stock

By default the model uses the net capital stock reported directly by Eurostat for the requested year. As an option, it instead constructs a perpetual-inventory series, $K_t = (1 - \delta)K_{t-1} + I_t$,

seeded from the earliest year with both a reported stock and investment series, with δ estimated per sector from the ratio of consumption of fixed capital to stock in that base year rather than assumed. This uses a single depreciation rate per sector, not differentiated by asset type; we return to this simplification in Section 6.2.

The aggregate AMECO capital-stock series used in Section 5.2 (item `OKND`, "net capital stock at 2020 prices") is, by contrast, already a constant-price (volume) series, whereas AMECO's wage (`UWCD`) and output (`UVGD`) series are current-price. The pipeline exposes both the raw combination and valuation-consistent alternatives that deflate the wage and output series by AMECO's own GDP deflator (`PVGD`=`UVGD`/`OVGD`×100) or investment-goods deflator (`PIGT`) onto the same basis as the capital stock; Section 5.2 reports all three.

To avoid the ambiguity of an unqualified "surplus," we distinguish

$$S^{\text{gross}} = Y - V, \qquad S^{\text{net}} = Y - V - \text{CFC}, \tag{12}$$

and state explicitly which is used where. The invariance argument above concerns S^{gross}: $(Y - V)/V$ is exactly invariant under a common deflator applied to Y and V, with no dependence on how CFC is measured. The rate of exploitation e and profit rate r reported in Section 5.2 and throughout the aggregate time series use S^{net} (equivalently, $e = (Y - V - \text{CFC})/V$), with CFC drawn from AMECO's own consumption-of-fixed-capital series (item `UKCT`, current-price, deflated consistently with V and Y in each convention) rather than an assumed fraction of the capital stock; an earlier version of this calculation substituted a flat 10% of the (constant-price) capital stock for CFC: a constant-price, unscaled term that does not move with the deflator applied to Y and V, which breaks the algebraic invariance of S^{net}/V specifically (there being no CFC-like term in $S^{\text{gross}}/V = (Y - V)/V$ for a mismeasured CFC to disturb) — an inconsistency we found via external review and verify is now resolved in Section 5.2, where we report that e^{net} (computed with S^{net}, CFC deflated consistently with Y and V) is invariant across conventions exactly, confirming the fix addresses the right term. The sectoral cross-section of Section 5.1 also uses S^{net} throughout (business income after consumption of fixed capital, Section 4.3).

Sectoral estimates of e (Section 5.1) impose a floor of 0.1: a small number of sectors have business income, net of consumption of fixed capital, close to zero or negative relative to compensation, which the model's positivity requirement on e (Section 3.1) does not allow. This is an imposed value, not a purely empirical one. The pipeline reports explicitly how many sectors hit it, and keeps the raw ratio alongside the imposed one, so that Table 2 can show both rather than leaving the floor invisible in the reported figures.

5 Results: Portugal

5.1 Sectoral profile, 2022

Throughout this section and Section 5.2, e and r are computed from national-accounts aggregates (Section 4.3) and should be read as a *national-accounts proxy* for the theoretical rate of exploitation and profit rate, not a direct measurement of either. Business income net of consumption of fixed capital, the numerator in our construction, is not a clean measure of surplus value: it is not adjusted for mixed income of the self-employed (allocated to neither V nor S here), taxes less subsidies on production, imputed rents, non-market output, or other national-accounts conventions that do not map cleanly onto Marxian categories. This matters most for exactly the sectors where these conventions bite hardest — real estate, agriculture, public administration, education, and health — and the real-estate outlier discussed below is one visible symptom of it, not the only one. We do not attempt gross/net or mixed-income-adjusted variants of the sectoral profile in this paper; doing so properly is future work.

Table 1 reports the calibrated rate of exploitation, organic composition, profit rate, and gross value added (GVA, sourced directly from the underlying Eurostat data, not reconstructed

from model parameters) for all 19 sections, ranked by profit rate. Sectoral GVA sums to ≈ 209.8 thousand M€, matching Portugal's 2022 national accounts total by construction — an accounting identity the pipeline checks and that an earlier, buggy version of this table violated by two orders of magnitude, because it reconstructed a stand-in for GVA as $C_i+V_i+e_iV_i$ (adding a capital *stock* to two flows, the same conflation discussed in Section 3.1) instead of reporting the real underlying series.

Three different summary statistics for the sectoral rates are worth distinguishing, since they answer different questions and can diverge substantially: the simple (unweighted) mean across the 19 sectors is $e \approx 0.89$, $k \approx 24.8$, $r \approx 10.0\%$; the GVA-weighted mean is $e \approx 1.45$, $k \approx 44.2$, $r \approx 10.7\%$; and the properly aggregated rate, $\sum_i S_i / \sum_i (C_i + V_i)$ (summing numerators and denominators separately before dividing, rather than averaging ratios), is $r \approx 6.1\%$. The gap between these is driven substantially by a single outlier, real estate activities ($k \approx 334$), plausibly reflecting the well-documented effect of imputed owner-occupied rents on this sector's national-accounts capital intensity (a compositional effect of the accounting convention, not something we have independently verified is the sole driver here); it should be treated with caution when interpreting any of the three summary figures. Of the 19 sectors, 4 hit the $e = 0.1$ floor described in Section 4.3 (business income net of depreciation close to zero or negative relative to compensation): an imposed value in those cases, not a purely empirical observation. Table 2 lists their raw ratios next to the imposed values. The floor matters little in aggregate: $r = \sum S / \sum (C + V)$ is 6.12% with it and 5.80% with the raw ratios. Excluding real estate, sectoral profit rates range from financial and insurance activities at the top ($r \approx 36\%$) to public administration and utilities near the bottom (r below 2%), a dispersion consistent with the classical expectation that market prices, not labor values, equalize profit rates across sectors of differing organic composition — which Section 5.3 examines directly.

Table 1: Sectoral calibration, Portugal, 2022 (real Eurostat data). GVA is the sector's real gross value added, not a quantity reconstructed from other columns.

Code	Sector	GVA (€bn)	e	k	r (%)
K	Financial and insurance activities	11.4	1.51	3.17	36.1
S	Other service activities	2.5	0.62	1.97	20.83
I	Accommodation and food service	13.0	0.86	3.51	19.07
G	Wholesale and retail trade	26.0	0.53	2.05	17.37
A	Agriculture, forestry and fishing	4.6	1.39	8.3	14.98
N	Administrative and support	9.5	0.33	1.28	14.37
F	Construction	10.0	0.33	1.61	12.65
M	Professional, scientific and technical	10.6	0.34	1.79	12.23
C	Manufacturing	30.4	0.53	3.91	10.75
B	Mining and quarrying	0.7	0.65	9.44	6.19
J	Information and communication	10.0	0.22	2.84	5.71
P	Education	11.2	0.1	1.08	4.8
Q	Human health and social work	14.4	0.1	1.93	3.41
H	Transportation and storage	9.8	0.23	6.33	3.13
L	Real estate activities	25.6	8.52	333.82	2.55
E	Water supply, sewerage, waste	2.0	0.34	13.78	2.3
R	Arts, entertainment and recreation	2.1	0.12	4.15	2.29
O	Public administration	14.1	0.1	7.45	1.18
D	Electricity, gas, steam	1.8	0.1	62.91	0.16

Table 2: Sectors at the $e = 0.1$ floor, Portugal, 2022. Raw e is business income net of consumption of fixed capital over compensation, before the floor; raw $r = e/(k+1)$ uses that raw value. Generated by `paper/generate_results.jl`.

Code	Sector	raw e	imposed e	raw r (%)	imposed r (%)
D	Electricity, gas, steam	-0.527	0.10	-0.82	0.16
O	Public administration	-0.026	0.10	-0.31	1.18
P	Education	-0.010	0.10	-0.48	4.80
Q	Human health and social work	0.088	0.10	3.01	3.41

5.2 Aggregate profit-rate trend, 1996–2022

Figure 1 shows the aggregate rate of exploitation, organic composition, and profit rate for Portugal over 1996–2022, calibrated entirely from real AMECO series, under two valuation conventions (Section 4.3): the solid line combines AMECO's current-price wage and output series with its constant-price capital-stock series without adjustment (*current-price V, GVA*); the dashed line first deflates the wage and output series onto the same constant-price basis as the capital stock (*deflated V, GVA*), using AMECO's own GDP deflator, with the surplus computed from real consumption-of-fixed-capital data (Section 4.3), not an assumed depreciation rate. Under both conventions the rate of exploitation moves identically, from 0.785 in 1996 to 0.739 in 2022 (ranging 0.68 to 0.91 over the period, dipping sharply in 2020) — *exactly* identically, as the algebra in Section 4.3 requires once Y, V, and the depreciation deduction are deflated by a common index, and as we verify directly (maximum difference across all 27 years: 0, to floating-point precision). This was not the case in an earlier, buggy version of this calibration, where the two conventions diverged sharply (0.21 → 0.61 vs. 0.58 → 0.57) — the divergence was itself the symptom of the flat-depreciation bug, not a property of the underlying valuation question, and disappears once the bug is fixed. What remains convention-sensitive is the profit rate, through the capital-advanced denominator $C + V$: under the unadjusted convention it rises modestly from 7.9% (1996) to 11.5% (2022), ranging 7.9–11.5% over the full period; under the GDP-deflated convention it moves from 12.6% to 10.9% at the endpoints but ranges 9.3–12.7% over the full period (the endpoints are not the extremes); under an investment-deflator convention (`PIGT` in place of `PVGD`, Section 6.1) from 15.0% to 12.3% at the endpoints, ranging 9.3–15.1% over the full period. Across all three conventions and all 27 years, the profit rate's global range is 7.9–15.1% — a real, moderate sensitivity to the deflator chosen, moving mildly within each convention rather than dramatically, and not the near-fivefold artifact the earlier, buggy calibration reported. Figure 2 repeats the GDP-deflated calculation for Spain, using the identical pipeline: the rate of exploitation again moves identically to the unadjusted convention (verified the same way), from 0.85 to 0.76, while the profit rate moves from 12.0% to 9.1% under the deflated convention (vs. 7.9% to 9.6% unadjusted) — the same qualitative picture as Portugal: a real but modest level of convention-sensitivity confined to the profit rate. We discuss the implications of this in Section 6.1.

5.3 The transformation problem on the real economy

Figure 4 compares labor values with Sraffian production prices computed on the real calibrated matrix, for three empirically plausible uniform profit rates ($\bar{r} = 5\%, 10\%, 15\%$). With the corrected technical matrix (Section 4.2), $R_{\max} \approx 91\%$ for this economy — still higher than any plausible real profit rate, but no longer the $\approx 691\%$ an earlier, buggy version of the matrix construction implied, and squarely in an economically interpretable range for a maximum feasible rate. We still plot at concrete empirical rates rather than fractions of $R_{\max}$, since $R_{\max}$ itself is not a rate any real economy operates at; we checked that the qualitative pattern below is

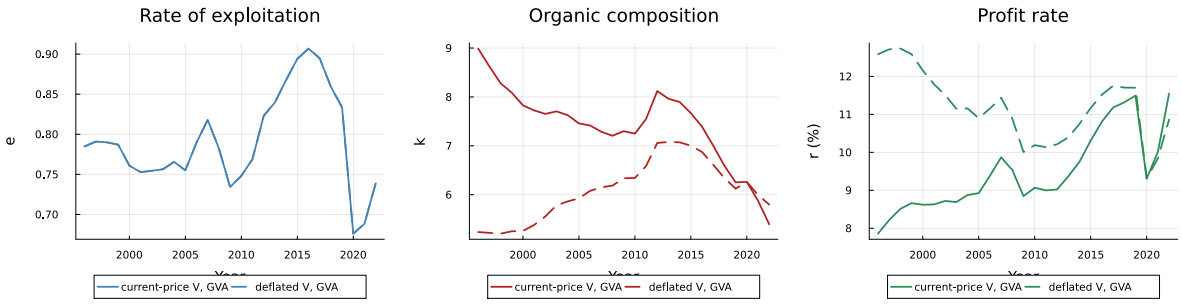


Figure 1: Rate of exploitation, organic composition, and profit rate, Portugal, 1996–2022 (real AMECO data), under current-price and valuation-consistent (deflated) conventions.

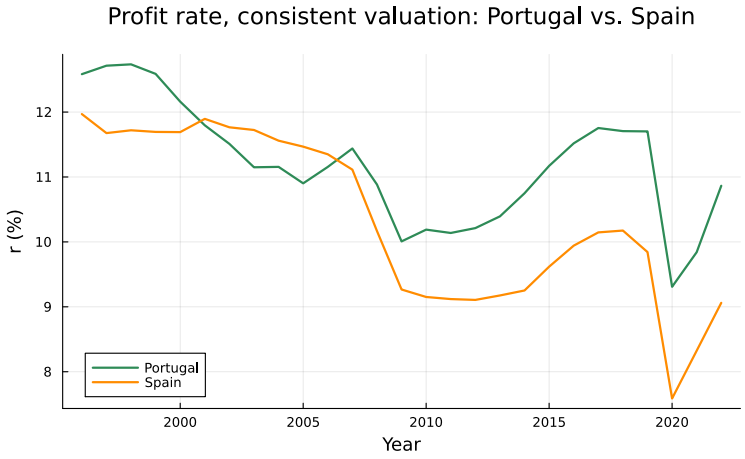


Figure 2: Profit rate under the GDP-deflated convention, Portugal vs. Spain, 1996–2022 (real AMECO data).

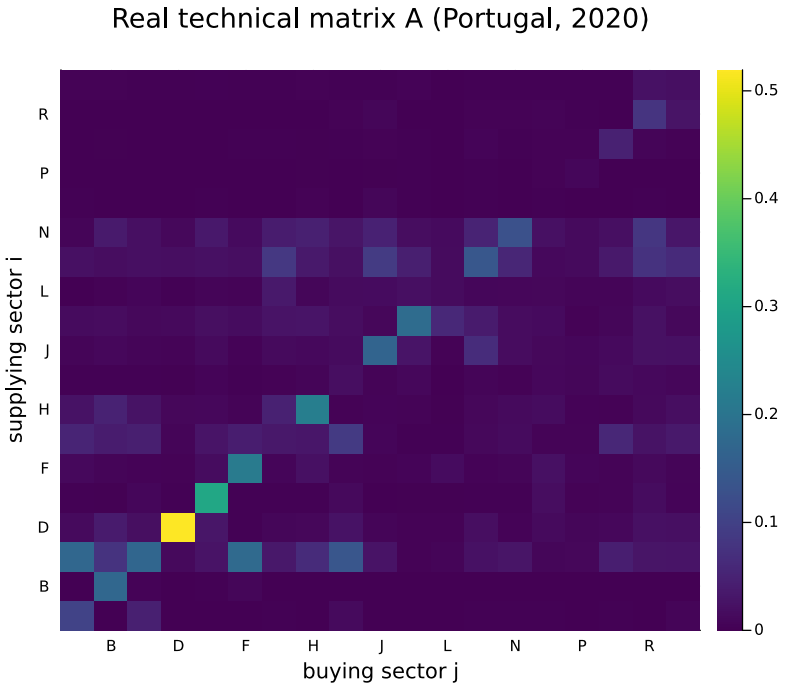


Figure 3: Real technical matrix A, Portugal, 2020 (19 CPA product sections; the benchmark input-output table's most recent available year, used coherently with 2020 labor coefficients — Section 4.2).

unchanged at $0.3R_{\max} \approx 27\%$ and $0.6R_{\max} \approx 54\%$ without including a second figure. As $\bar{r}$ rises, sectors with above-average organic composition see their production price rise relative to their labor value, and vice versa, the qualitative signature the transformation problem predicts, now shown on a real rather than a randomly generated technical structure. The comparison is qualitative only: labor values are expressed in labor-time units and production prices in the wage-numeraire monetary units of Equation (3), and we do not fix an explicit monetary expression of labor time (MELT) to convert between them. Figure 4 therefore reports each series normalized to sum to one and compares relative deviations, which is sufficient to display the sign and ordering of the transformation effect but does not license a literal price-value comparison in a common unit — a well-known difficulty in empirical Marxian value theory [Freeman and Carchedi, 1996, Basu and Manolakos, 2013].

5.4 Comparison of coupling mechanisms: one illustration, then a systematic sweep

Because no real per-sector accumulation time series exists to calibrate the dynamic coupling mechanisms of Section 3.3 empirically, Figure 5 first illustrates their qualitative behavior on a single synthetic six-sector economy under one random technical matrix and seed. Capital mobility alone reduces cross-sector profit-rate dispersion over time, as expected. Adding the aggregate realization-pressure term leaves the qualitative picture largely unchanged in this run. Adding exact quantity rationing instead *increases* final dispersion relative to mobility alone in this particular random economy: rationing can prevent the mobility term from fully equalizing rates when a high-profit sector's own suppliers cannot support the investment mobility would otherwise direct there.

An earlier draft reported this as an example of the mechanism's behavior without evidence it generalizes beyond one random seed. We checked directly rather than leaving that as an open question: Figure 6 repeats the same three-way comparison across 50 independent random



Figure 4: Labor values vs. Sraffian production prices, real technical matrix and labor coefficients, Portugal 2020 (a coherent single-year calibration — Section 4.2), at three illustrative profit rates.

seeds (each with its own random six-sector technical matrix, sectoral parameters, and initial conditions). Rationing produces *higher* final dispersion than mobility alone in **50 of 50** seeds — the pattern in Figure 5 is not an artifact of one unlucky draw, but a consistent finding across this specific class of random six-sector economies (moderate density, the parameter ranges of `create_technical_matrix` and `create_departments`); we do not claim it generalizes beyond that tested class. The mean final dispersion is 0.063 (mobility only), 0.189 (adding realization pressure), and 0.448 (adding rationing) — rationing roughly triples dispersion relative to realization pressure and roughly seven-folds it relative to mobility alone, on average, with substantial spread across seeds (rationing's dispersion ranges 0.168–0.687 across the 50 economies). The economic reading is the same as before, now with numerical backing: exact per-sector rationing is a genuinely different, more disequalizing mechanism than either capital mobility or an aggregate realization heuristic, because it can trap a high-profit sector behind a supply-constrained supplier in a way neither of the other two mechanisms can represent.

We separately checked the rationing fixed point's convergence claim (Section 3.3) far beyond the closed-form example and the earlier twenty-parameter-combination sweep: 270 trials spanning $N \in \{6, 12, 19\}$ and matrix density $\in \{0.2, 0.4, 0.6\}$ (30 seeds per combination), with supply deliberately scaled to force genuine rationing rather than the trivial $\kappa \equiv 1$ case (268/270 trials had at least one rationed sector). The damped iteration converged in **270 of 270** trials, in a mean of 18.0 and a maximum of 20 iterations at the default $\alpha = 2/3$ (against a cap of 200; 25.8 and 30 at the previous default $\alpha = 0.5$, Table 3) — no failures, and no case requiring anywhere near the iteration budget. Proposition 1 now explains this local behavior for every admissible instance (Section 5.5 checks it against these same 270 economies); global convergence from arbitrary initial ρ and uniqueness remain empirical (Section 3.3).

5.5 Exact Jacobian, bifurcations, and TSSI valuation

The rationing spectrum. We recomputed the exact Jacobian of the rationing map at the fixed point of each of the same 270 economies of Section 5.4 (same seeds), using automatic differentiation rather than the hand formula. Of the 268 with genuine rationing, all 268 have a Jacobian that is lower triangular once sectors are ordered by ρ^*, with eigenvalues equal to its diagonal; all 268 are antitone (every entry ≤ 0); 0 have a complex eigenvalue; and in all 268 the most rationed sector's row is exactly $-e_{k_0}^\top$, so $\lambda = -1$ is an eigenvalue (Figure 8, left). The local stability bound

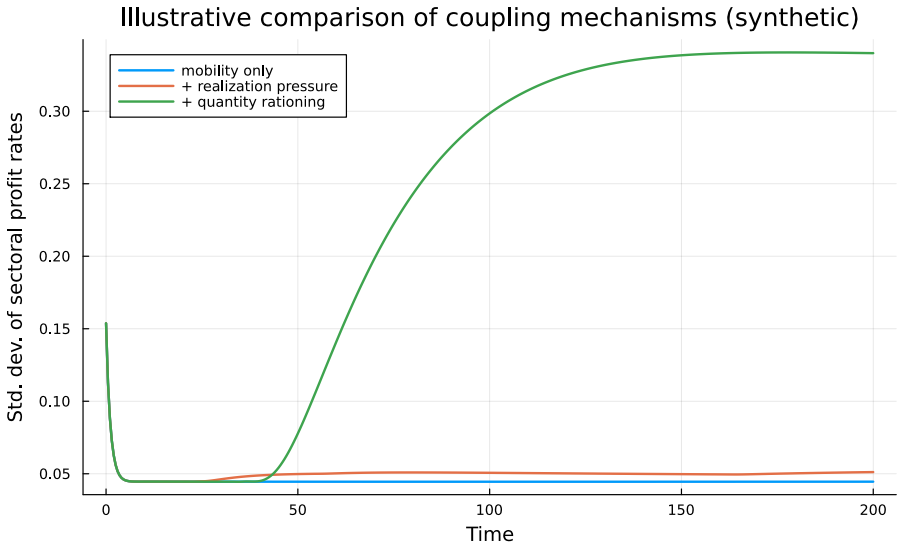


Figure 5: Cross-sector profit-rate dispersion over time under three coupling mechanisms, one illustrative synthetic economy (not calibrated to real data).

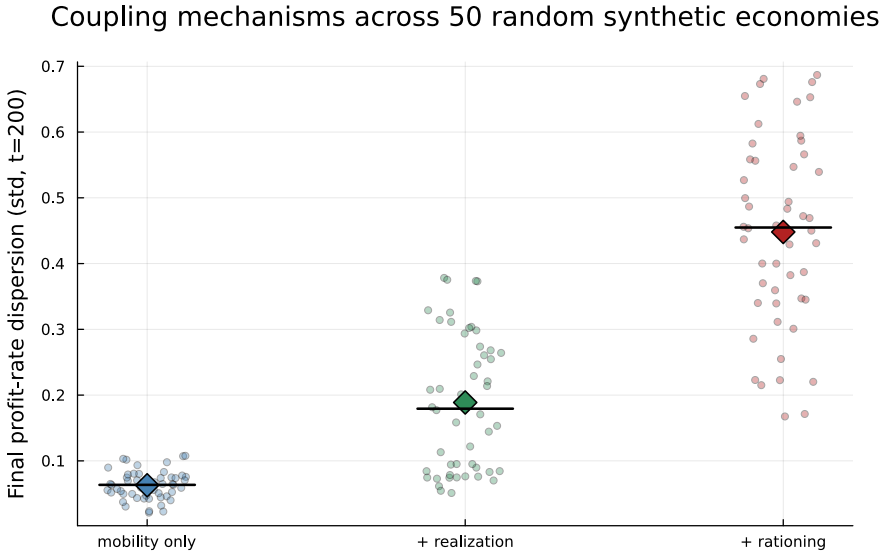


Figure 6: Final profit-rate dispersion under three coupling mechanisms, across 50 independent random synthetic economies (points: individual seeds, jittered; diamond: mean; bar: median).

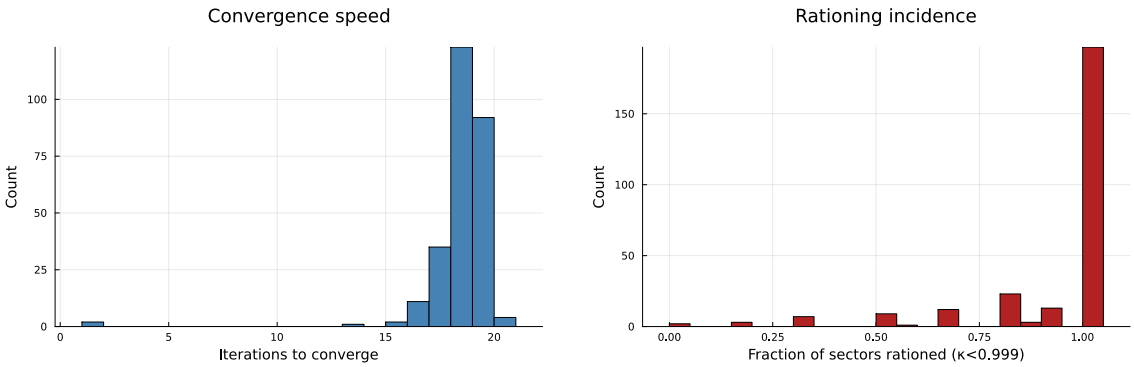


Figure 7: Rationing fixed-point convergence sweep, 270 trials ($N \in \{6, 12, 19\}$, density $\in \{0.2, 0.4, 0.6\}$, 30 seeds each). Left: iterations to converge (all 270 trials converged). Right: fraction of sectors genuinely rationed per trial.

is $\alpha_{\max} = 1.0$ in all 268 cases. Every rationed economy also has an eigenvalue 0 (268 of 268), so the radius-minimizing relaxation is $\alpha = 0.667$ in every case, and the spectral radius at the default $\alpha = 2/3$ never exceeds 0.333 (against 0.5 at the previous default $\alpha = 0.5$), exactly as Proposition 1 predicts. The radius predicts the iteration count: contracting the error by a factor 1/3 per step from an $O(1)$ start reaches the 10^{-8} step tolerance in $\log 10^{-8}/\log(1/3) \approx 16.8$ iterations, against an observed mean of 18.1 and maximum of 20; at $\alpha = 0.5$ the prediction is $\log 10^{-8}/\log(1/2) \approx 26.6$, against an observed 26.0 and 30 (Figure 8, right).

Proposition 1 guarantees only local convergence, so before changing the default we checked whether $\alpha = 2/3$ loses robustness far from ρ^*, where a smaller α might seem to keep a safer margin from the $\alpha_{\max} = 1$ boundary. It does not (Table 3): across the 270 sweep economies started at $\rho_0 = 1$, the same economies from 20 random initial ρ_0 each, and 200 further economies with far scarcer supply (N up to 30, each from $\rho_0 = 1$ and five random starts) — 6870 runs per value of α — every relaxation from 0.4 to 0.95 converged in every run (0 failures in total). Iteration counts follow the local prediction closely everywhere: they are minimized at $\alpha = 2/3$, about 30% below $\alpha = 0.5$, and rise slowly below the optimum but steeply above it, as the multiplier $1 - 2\alpha$ of the $\theta = 1$ mode approaches -1. We therefore made $\alpha = 2/3$ the package default. The fixed point itself does not depend on α: along the trajectory of Figure 5 the rationing factors computed with $\alpha = 2/3$ and $\alpha = 0.5$ differ by at most 2.4×10^{-9}, within the 10^{-8} solver tolerance, so no dynamic result in this paper changes.

Table 3: Iterations to converge (tolerance 10^{-8}) of the damped rationing iteration by relaxation α. Every run converged.

	Sweep ($\rho_0 = 1$)		Random ρ_0		Scarce supply	
α	mean	max	mean	max	mean	max
0.4	34.3	40	36.5	42	37.2	42
0.5 (previous)	25.8	30	27.8	32	28.2	32
0.6	20.5	24	21.7	25	22.0	25
2/3 (default)	18.0	20	18.6	22	18.8	21
0.75	25.8	28	26.3	29	25.8	29
0.85	50.2	53	50.7	54	49.0	54
0.95	174.4	180	169.2	181	165.3	181

Bifurcation diagram. Figure 9 continues the rationing fixed point of a 12-sector economy in the demand scale s. The analytic onset is $s_c = 0.117$. The stability boundary $\alpha_{\max}(s)$ is 2 below s_c and drops to 1 at it, staying there as the fraction of rationed sectors rises to one: the default

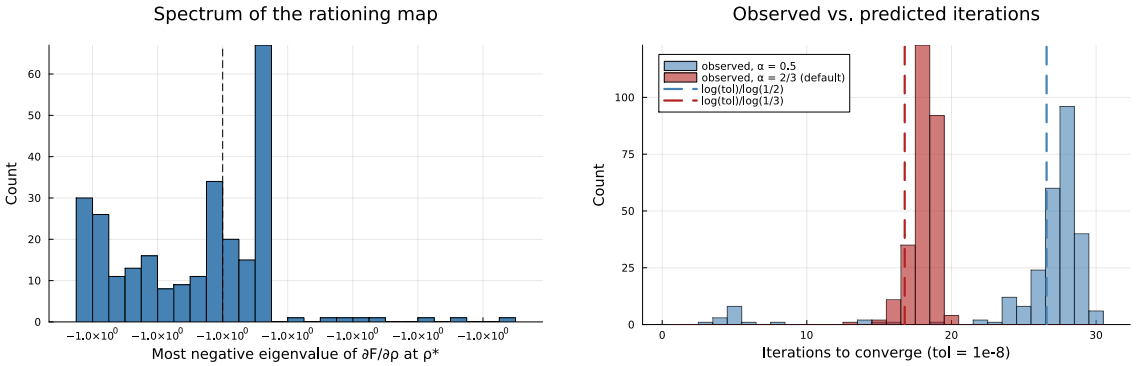


Figure 8: Exact (automatic-differentiation) Jacobian of the rationing map across the 268 rationed economies of the Section 5.4 sweep. Left: most negative eigenvalue, -1 in every case. Right: iterations to converge at the default $\alpha = 2/3$ and the previous $\alpha = 0.5$, against the $\log(\text{tol})/\log \varrho$ predictions from the exact spectrum ($\varrho = 1/3$ and $1/2$).

$\alpha = 2/3$ is stable along the whole branch (0 bifurcation points detected), while continuing the same branch with $\alpha = 1.1$ detects a flip at $s = 0.125$, the first continuation step past s_c — the border-collision flip of Section 3.5, located to the resolution of the continuation step.

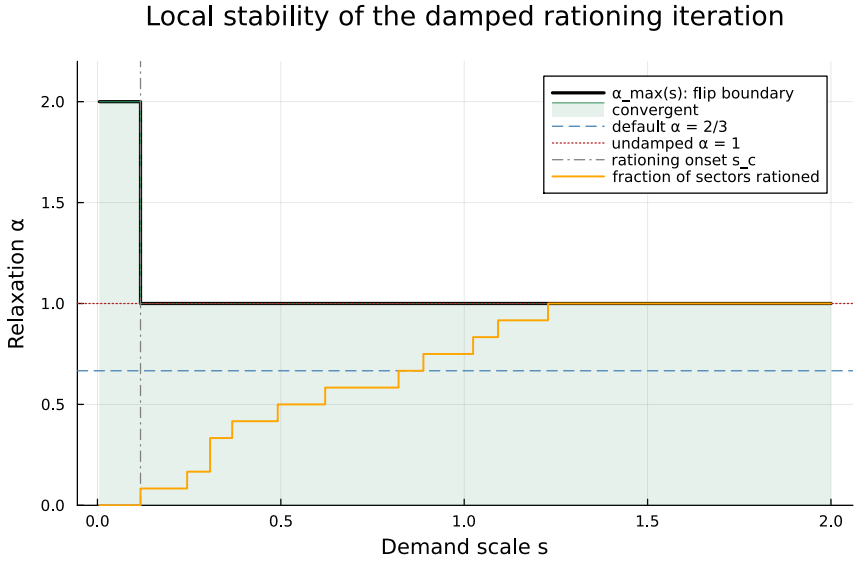


Figure 9: Local stability region of the damped rationing iteration in the (demand scale s, relaxation α) plane, from numerical continuation of the fixed point with exact Jacobians. The flip boundary $\alpha_{\max}(s)$ jumps from 2 to 1 at the border-collision onset of rationing s_c.

TSSI valuation. Along a synthetic six-sector trajectory with capital mobility and realization pressure, the TSSI and simultaneist profit rates stay close. After an initial transient in which TSSI lies below the simultaneist rate (inputs inherited from the initialization), the gap changes sign and widens with the lag: $r^{\mathrm{T}} - r^* = 0.011$ percentage points at the end of the run for $\tau = 1$, and 0.057 for $\tau = 5$ (Figure 10, left). The gap is small here because the technical matrix is fixed: with no technical change, input prices move only through the slowly changing cost structure, so historical and current input prices barely differ. With frozen data, the TSSI price map is locally stable in all 50 random economies tested (spectral radius of Equation (9) with mean 0.216 and maximum 0.542; Figure 10, right), so TSSI prices converge to simultaneist ones, with a half-life of about 0.45τ at the mean radius. Continuing each economy's fixed point in the

surplus-value scale $\vartheta \in [0.05, 10]$ with $\tau = 1$ fixed detects 0 bifurcations; the largest spectral radius reached anywhere on these branches is 0.967, approached as r^* nears $R_{\max}$. Within this class of economies, then, divergence between TSSI and simultaneist valuations requires changing data — accumulation or technical change — not an instability of the temporal price map itself.

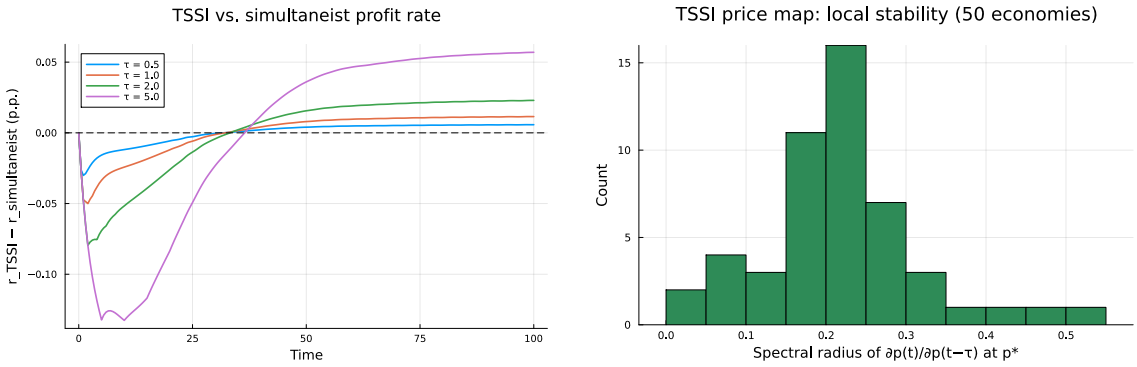


Figure 10: TSSI valuation. Left: TSSI minus simultaneist profit rate along one synthetic trajectory, for four production lags τ. Right: spectral radius of the TSSI price map at the simultaneist fixed point, 50 random economies (stable below 1).

5.6 Balanced growth, the wage share, and accumulation cycles

All results in this subsection are for the cross-dual model of Section 3.6 on synthetic economies, generated by `paper/accumulation_bifurcation.jl`. The reference economy has 6 sectors and capital mobility $\mu = 0.5$. Along its balanced-growth ray, the level system of Equation (11) grows at the predicted $g^\star = 24.4\%$ (relative error 4.4×10^{-13} over a simulation of 30 periods), and the TSSI and simultaneist profit rates of Section 3.4 coincide to within 9.8×10^{-14}: on a balanced-growth path the per-unit data are constant, so the TSSI price map sits at its simultaneist fixed point.

Distribution. With variable-capital mobility switched off ($\mu_v = 0$) and the default price-adjustment speed $\beta = 0.5$, balanced growth is locally stable at every one of 113 points of the two wage-share sweeps (uniform $\omega \in [0.15, 0.7]$, and proportional scaling of the sectoral rates of exploitation, reaching $\omega \in [0.2, 0.78]$). The leading eigenvalue's real part stays between -0.14 and -0.12. The growth rate falls from 102% to 7.4% and the aggregate profit rate from 141% to 14.2% as ω rises; the top of that range corresponds to implausibly low wage shares and serves only to show the shape of the relation. The aggregate organic composition falls from 3.02 to 2.01. The two sweeps fall on the same $g^\star(\omega)$ and $r^\star(\omega)$ curves (Figure 13, left): in this model, distribution enters balanced growth essentially through the aggregate wage share alone, not its sectoral pattern. The ratio $g^\star/r^\star$ declines from 0.72 to 0.52 across the sweep, a Cambridge-type relation whose effective saving rate falls as depreciation weighs more. The approach to balanced growth is oscillatory (the leading eigenvalue is complex) except for $\omega \geq 0.66$.

Hopf bifurcation in μ_v. With slower price adjustment, $\beta = 0.1$, balanced growth is stable at $\mu_v = 0$ (leading real part -0.031) and loses stability at $\mu_v = 1.003$, where a complex pair crosses the imaginary axis with speed $\mathrm{Re}\, a = 0.092$ and frequency 0.301 (period 20.9). The first Lyapunov coefficient is $\ell_1 = -0.193 < 0$: the bifurcation is supercritical, and a small stable accumulation cycle is born. The continuation finds 5 Hopf points on the equilibrium branch, the last at $\mu_v = 1.41$; only the first changes the stability of balanced growth, the others add further unstable directions. Figure 11 shows the stability of balanced growth and the branch of limit cycles continued from the first Hopf point. The cycle grows quickly in amplitude (measured as the largest swing of any sector's share of total capital) and stays stable up to a Neimark–Sacker bifurcation at $\mu_v = 1.206$, where it gives way to an invariant torus. The branch then turns back

at $\mu_v \approx 1.48$ and forward again at a fold of cycles at $\mu_v = 1.37$; shortly after, the cycle, now of shorter period, regains stability. The upper turning point is not flagged by the Floquet-based detector at the continuation step used, a reminder that bifurcations along the cycle branch are step-dependent. Further Neimark–Sacker points (at $\mu_v = 1.82$, 2.47) and a period doubling at $\mu_v = 2.07$ follow. By $\mu_v = 2.5$ the cycle has period 8.5 and amplitude 0.47, and the smallest sector share along it falls to 3.5×10^{-4}: the oscillation nearly wipes out sectors in turn, an economically implausible feature that we return to in Section 6.2.

Direct simulation on a grid of 20 values of μ_v confirms this picture (diamonds in Figure 11, right). Simulations converge to balanced growth below the Hopf point. They settle on a periodic orbit in 11 cases, 10 of which lie on a stable part of the continued branch, with periods within 1.0% and amplitudes within 3.9% of the collocation values (for example, period 19.3 at $\mu_v = 1.1$). The remaining periodic case ($\mu_v = 2.0$, period 8.7) lies where the continued cycle is unstable; we read it as phase locking on the torus rather than as a point of the branch. The 7 irregular cases — quasi-periodic between the first Neimark–Sacker point and the fold, and irregular from $\mu_v \approx 1.9$ — are exactly where the continued cycle is unstable.

Coexistence and chaos. The fold of cycles implies that a stable cycle and the torus may coexist. Sweeping μ_v up and then down by simulation in steps of 0.01 confirms it (Figure 12, right). For $\mu_v \in [1.37, 1.43]$ the economy ends on the torus (mean amplitude 0.59) when μ_v is raised and on a stable cycle (mean amplitude 0.51, period 13.9 falling to 13.1) when it is lowered: which accumulation regime prevails depends on the path by which mobility reached its current value. The lower edge of the window is the fold of cycles; the upper edge is not the branch's turning point at $\mu_v \approx 1.48$ but the destruction of the torus, between $\mu_v = 1.43$ and 1.44.

The three largest Lyapunov exponents, estimated over 30000 time units after a transient (Figure 12, left), separate the regimes. The control cases come out as theory requires: on the stable cycle at $\mu_v = 1.1$, $|\lambda_1| = 1.2 \times 10^{-5}$ and $\lambda_2 = -0.0067$; on the torus at $\mu_v = 1.3$, two exponents vanish ($|\lambda_1| = 7.6 \times 10^{-5}$, $|\lambda_2| = 4.2 \times 10^{-5}$) and $\lambda_3 = -0.0093$. Beyond the second Neimark–Sacker point the largest exponent is positive at $\mu_v = 1.9$, 2.1, 2.2, 2.3, 2.4 (λ_1 between 0.028 and 0.097), and it agrees within 6% between the default and a randomly perturbed initial condition. At $\mu_v = 2.0$ it is zero to within 4.9×10^{-5}, a periodic window corresponding to the phase-locked orbit found above. At $\mu_v = 2.2$, 2.3, 2.4 a second exponent is also positive (between 0.009 and 0.013 across both initial conditions), the signature of hyperchaos. The irregular regime is therefore chaotic, by the standard numerical criterion. This is evidence, not proof: the exponents are finite-time estimates, and the chaotic trajectories, like the cycles, come close to the boundary of the simplex (smallest sector share 7.4×10^{-4}).

Hopf boundary. Repeating the continuation for uniform wage shares $\omega \in [0.2, 0.65]$ traces the Hopf boundary $\mu_{v,H}(\omega)$ (Figure 13, right). The boundary is supercritical throughout and rises with the wage share, from 0.76 to 1.42: a higher wage share makes balanced growth more robust to variable-capital mobility. Between $\omega = 0.25$ and 0.3 the frequency at the boundary jumps from 0.47 to 0.34: a different eigenvalue pair becomes the first to cross, so a double-Hopf point lies in that interval. Faster price adjustment is strongly stabilizing: for $\beta \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1.0\}$ the first Hopf point lies at $\mu_v = 0.89$, 1.0, 1.21, 1.42, 1.84, 2.91 respectively.

Robustness. In 30 random economies with $N = 3$–8 sectors and $\beta = 0.1$, balanced growth at $\mu_v = 0$ is stable in 30, loses stability at a Hopf point in $\mu_v \in [0, 4]$ in 30, and that first Hopf point is supercritical in 30. It lies between $\mu_v = 0.55$ and 1.29 (median 0.91), with periods of the bifurcating cycle between 14.2 and 30.4; no sector share on the equilibrium branches falls below 0.019. The supercritical Hopf bifurcation to accumulation cycles is therefore generic in this class of economies, not a feature of the reference example.

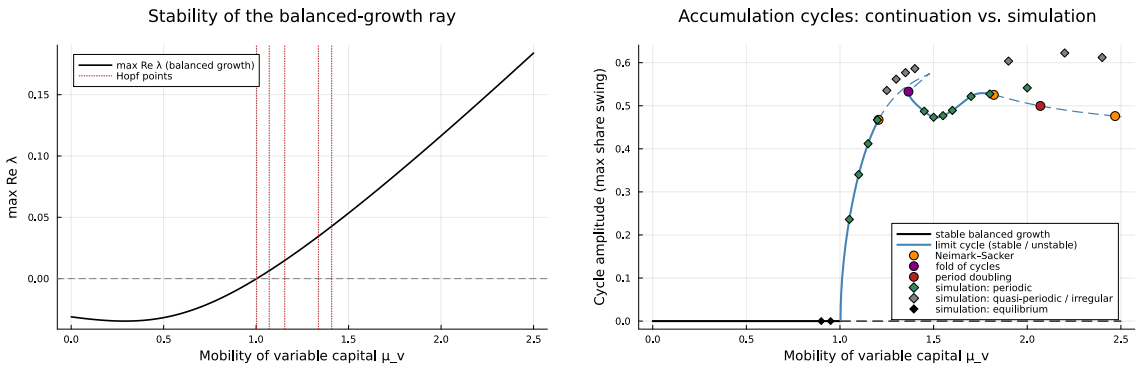


Figure 11: Accumulation cycles in the cross-dual model ($N = 6$, $\beta = 0.1$). Left: largest real part of the eigenvalues of the normalized system along the balanced-growth branch, with the Hopf points. Right: amplitude of the limit cycles continued from the first Hopf point (solid: stable; dashed: unstable), with their bifurcations, against direct simulations classified by their Poincaré section.

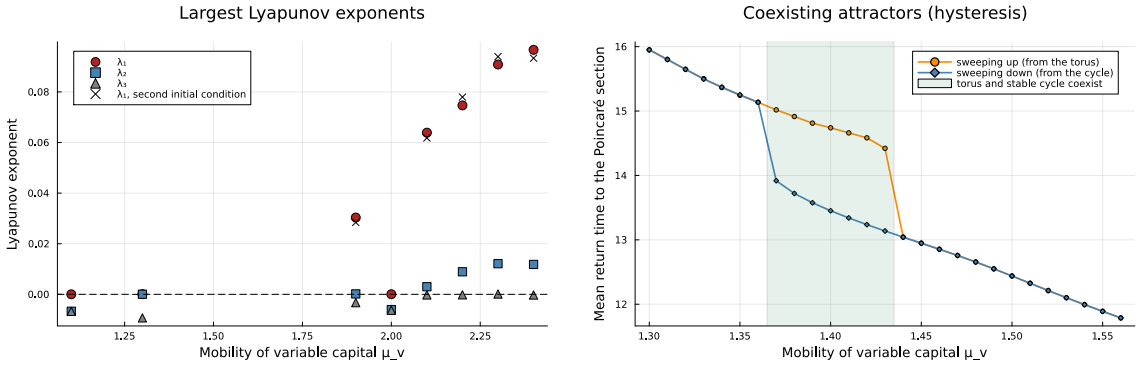


Figure 12: Chaos and coexisting attractors in the reference economy ($N = 6$, $\beta = 0.1$). Left: three largest Lyapunov exponents of the normalized system ($\times$: λ_1 from a second initial condition). Right: mean return time to a Poincaré section when μ_v is swept up from the torus and down from the stable cycle; in the shaded window both attractors coexist.

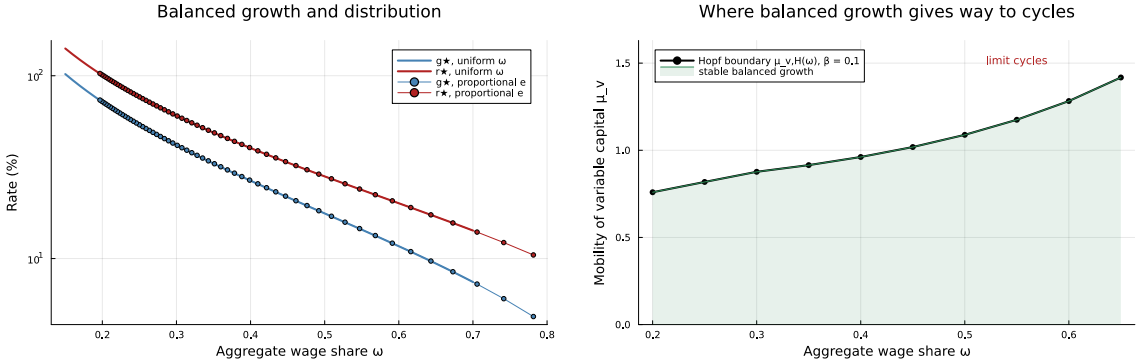


Figure 13: Distribution and balanced growth. Left: growth and aggregate profit rates on the balanced-growth ray against the wage share, for a uniform wage share and for a proportional scaling of sectoral exploitation ($\mu_v = 0$, $\beta = 0.5$). Right: Hopf boundary in the (wage share, variable-capital mobility) plane ($\beta = 0.1$).

6 Discussion

6.1 A stock-flow conflation and a valuation artifact

A first pass at the raw AMECO series appears to show a profit rate rising sharply for Portugal over 1996–2022, driven by an organic composition that falls even as the rate of exploitation nearly triples. This is the pattern we originally reported. It does not survive a closer look, and tracing why it does not survive turned out to require finding two separate defects, not one.

The first defect was in the model, not the data. Gross output throughout MarxKapital was computed as $O_i = C_i + V_i + S_i$ (Section 3.3), adding a capital *stock* to two annual *flows*; the aggregate AMECO calculation used the same conflation implicitly, deducting a flat 10% of the (constant-price) capital stock as "depreciation" from a current-price surplus. External review of an earlier draft caught this. It matters here specifically because the flat-depreciation term breaks an algebraic identity that should otherwise hold for the quantity actually reported. The rate of exploitation used throughout this paper is net of consumption of fixed capital, $e^{\text{net}} = (Y - V - \text{CFC})/V$ (Section 4.3); if Y, V, and CFC are *all* deflated by a common price index, e^{net} is invariant under that deflation — the common factor cancels from every term. The earlier, buggy version deducted a flat 10% of the (constant-price) capital stock in place of CFC: an unscaled term that does *not* move with the deflator applied to Y and V, so it breaks precisely the invariance e^{net} should have, and breaks it in a way that depends on how much the deflator itself has moved — which changes over 26 years of cumulative inflation, producing not just a level shift between conventions but a diverging *trend*. This, not a genuine valuation difference, is what generated the original $e : 0.21 \to 0.61$ figure. (The purely gross quantity $S^{\text{gross}}/V = (Y - V)/V$, which never involved CFC at all, was never the source of the problem; it is trivially invariant under any common deflator regardless of this bug, since no unscaled term ever entered it.)

We fixed the underlying conflation directly (Section 3.1, Section 4.3): output-facing formulas now use $\delta_i C_i$ (capital *transferred* in a period) rather than C_i (capital *advanced*, a stock), and the aggregate calculation now uses AMECO's own consumption-of-fixed-capital series (properly deflated alongside Y and V) rather than an assumed rate. The predicted consequence is checkable, and we checked it directly rather than asserting it: with the fix, e^{net} is now *exactly* identical between the unadjusted and GDP-deflated conventions for every year in the 1996–2022 series, for both Portugal and Spain (Section 5.2). This is not a small improvement in agreement — it is exact equality to floating-point precision, which is what the algebra predicts once every term shares a common deflation factor, and is the strongest kind of internal-consistency check a reader could ask for: not "the numbers look more reasonable now," but "the two conventions are now provably required to agree, and they do."

The second, genuine question is the one we originally set out to answer: does AMECO's capital stock (constant-price) versus wage and output series (current-price) distort the measured profit rate once the depreciation bug is out of the way? It does, but modestly. Portugal's profit rate now ranges 7.9%–15.1% across three deflation conventions over the full period (Section 5.2) — rising mildly from 7.9% to 11.5% unadjusted, declining mildly at the endpoints from 12.6% to 10.9% (ranging 9.3–12.7% across the full period) under the GDP deflator, and from 15.0% to 12.3% at the endpoints (ranging 9.3–15.1%) under an investment-goods deflator. We checked the investment-deflator variant specifically because AMECO's capital stock is a chain-linked series built from investment flows using an investment-specific price index (`PIGT`), not the general GDP deflator (`PVGD`); the two give different *levels* but the same qualitative shape (mild decline), evidence the residual sensitivity is not an artifact of which deflator was picked. Applying the GDP-deflated convention to Spain (Figure 2) shows the same modest pattern: a profit rate moving from 12.0% to 9.1% against 7.9% to 9.6% unadjusted. We do not read either country's series as proof of a general tendential pattern from a two-country, single-source comparison; national-accounts profit-rate estimates remain sensitive to depreciation conventions and sectoral

composition, discussed further in Section 6.2. We also could not find, and do not believe AMECO publishes, a current-price (replacement-cost) capital-stock series that would let us skip deflation entirely and compare all three quantities at a common current-year valuation, which would be the cleanest check in principle.

We report the full sequence — the original artifact, the stock-flow bug that substantially manufactured it, the fix, and the smaller genuine sensitivity that remains — rather than only the corrected numbers, because the sequence is the more durable methodological finding here: an automated pipeline that conflates a stock with a flow, or mixes valuation conventions across variables pulled from the same database, can each independently manufacture a statistically salient but spurious trend, and the two can compound. Disaggregating the corrected aggregate trend by sector, and comparing it against the 2022 cross-section of Table 1, is beyond the scope of this paper — complicated further by the single-year nature of the sectoral calibration versus the time-series nature of the aggregate result — and is left for future work.

6.2 Limitations

We report limitations transparently, following the same practice as the package's own documentation.

Default behavior remains synthetic. Both calibration paths default to synthetic data or fixed sectoral weights; every real-data result in this paper required explicit opt-in flags. This is a deliberate design choice to keep the package's automated test suite fully offline, not a limitation of the real-data pipeline itself, but it means synthetic results should never be mistaken for empirical ones when using the package's defaults.

Single-rate depreciation. The perpetual-inventory capital-stock option (Section 4.3) uses one depreciation rate per sector, not differentiated by asset type. Eurostat publishes investment by asset type for the relevant dataflow, but not the corresponding asset-specific depreciation profiles, which would be needed for a fully differentiated perpetual-inventory calculation.

The dynamical model's output measure was resolved; final demand was not. An earlier version of this paper used a value-added-type flow $W_i = \delta_i C_i + V_i + S_i$ as the supply side of Section 3.3's coupling mechanisms, omitting circulating constant capital (materials, energy, purchased services) entirely. We closed this gap rather than continuing to disclose it: Equation (2) reconstructs genuine gross output x_i from W_i and the model's own cost structure at every solver evaluation, without adding new state, and both coupling mechanisms now use x_i. The investment-demand side was fixed correspondingly (Equation (5)): A_{ij} now multiplies a desired *output* expansion Δx_j, converted from planned investment expenditure via the sector's own capital-output ratio, not raw expenditure directly. One gap remains, disclosed rather than resolved: the quantity-rationing mechanism still treats the *full* reconstructed x_i as available to meet investment demand, though output must also replace intermediate inputs and meet consumption and other final demand first. Closing this fully would require an explicit final-demand allocation across sectors, for which no data source in the current pipeline provides sector-of-origin consumption baskets; until then, $D_i \leq x_i$ should be read as an upper bound on investable supply, not a fully netted reproduction balance. This limitation is narrower than before — confined to the investable-residual question, not the output measure itself.

Limited robustness and sensitivity analysis, partially addressed. We report the aggregate trend under three valuation conventions (unadjusted, GDP-deflated, investment-deflated; Section 5.2 and Section 6.1) and for two countries (Portugal and Spain), which is enough to show the corrected finding is not a single-country, single-deflator artifact, but not enough to establish a general pattern across peripheral EU economies, nor to rule out sectoral composition effects that an aggregate series cannot separate from a genuine average-rate change. The sectoral profile (Table 1) still uses a single set of choices: net capital stock rather than perpetual-inventory, and a single value of the mobility coefficient μ. For the coupling mechanisms of Section 3.3, we no longer rely on a single random seed: a 270-trial sweep (Section 5.4) confirms the rationing

fixed point converges reliably (270/270) across varied N, matrix density, and demand/supply configurations, and a 50-seed sweep confirms rationing produces higher dispersion than mobility alone systematically (50/50 seeds), not as an artifact of one draw. What these sweeps do not cover: a fixed μ and fixed relaxation parameter throughout, and only random (not empirically calibrated) technical matrices for the dynamic comparison, since no real per-sector accumulation series exists to calibrate it against (Section 5.4). Section 3.3 establishes that a fixed point always exists (Brouwer), explains structurally why undamped iteration oscillates rather than converges (the map is antitone, and Proposition 1 shows undamped iteration sits exactly on a flip boundary whenever rationing binds), gives direct empirical evidence of practical uniqueness under variation in initial conditions, sector ordering, and tolerance in one genuinely-rationed test case, and proves that damped iteration converges locally, at a rate the exact spectrum predicts, for every relaxation $\alpha \in (0, 1)$ at every generic fixed point. What remains open is a uniqueness *proof* and a *global* convergence guarantee from arbitrary initial ρ across all admissible (A, σ, S, O) — and Section 3.3's own negative result suggests the obvious route to the latter, a global Lipschitz bound in the spirit of $\rho(A) < 1$, cannot supply it: we showed algebraically that its tightest admissible form always exceeds 1 whenever any sector is actually rationed. Closing this gap, if it can be closed at all, likely requires a different technique — possibly exploiting the antitone structure directly (e.g. bounding $F \circ F$) rather than a Banach-style argument on F itself — and we continue to rely on the empirical evidence above for the general case in the meantime.

The technical matrix construction was audited, and the audit found a bigger problem than expected. We set out to quantify a margin-clipping step's effect, reconcile row/column totals, document the CPA-to-NACE concordance, and resolve the domestic-vs-total-use question (`paper/matrix_audit.jl`). Instead we found the pipeline had been pulling from the wrong Eurostat dataflow entirely (Section 4.2): `NAIO_10_CP1620` is a trade-and-transport-margins satellite table, not the symmetric input-output table, and building A from it understated intermediate flows by more than an order of magnitude — the direct cause of the implausibly low $\rho(A) \approx 0.13$ and implausibly high $R_{\max} \approx 691\%$ this paper previously reported, and of every negative cell we had attributed to a margin-accounting convention. Rebuilt from the correct table (`NAIO_10_CP1700`), the matrix has zero negative cells, reconciles *exactly* against the table's own published output row for all 19 sections, and gives $\rho(A) \approx 0.52$, $R_{\max} \approx 91\%$ — both economically plausible. The domestic-vs-total-use question is also resolved rather than left open: the correct table publishes a genuine domestic-use variant, which we use, directly addressing the concern that imported inputs were being modeled as domestic supply. The CPA-to-NACE concordance question does not require a documented mapping *inside* the matrix construction step itself: the symmetric table's two dimensions (`prd_ava`, `prd_use`) both use the same CPA product code, so A is built entirely within one classification, with no product-to-industry join. A concordance assumption is still made one level up, where A's CPA-labeled sections are combined with NACE-classified labor coefficients (`NAMA_10_A64`) for the production-structure analysis of Figures 3–4. Table 1 is constructed independently, directly from NACE-classified sectoral accounts, and does not depend on A at all (Section 5.1). We rely on Eurostat's standard design correspondence between CPA and NACE section codes at this level of aggregation (both derive from the same top-level classification structure) rather than an independently verified concordance table, and label A's axes as CPA product sections, not NACE industry sections, to keep this distinction visible. What remains unresolved, and cannot be fixed by writing more code: `NAIO_10_CP1700` is a benchmark table published for specific reference years (2010, 2013, 2015, 2017, 2020 for Portugal), not annually. We resolved the resulting vintage question by scoping rather than by leaving a silent mismatch: Figures 3 and 4 now use a fully coherent 2020 calibration (matrix and labor coefficients both 2020), while Table 1 and the aggregate AMECO series remain genuinely 2022, since neither depends on A. The paper's empirical results therefore span two internally-coherent reference points rather than one section silently mixing 2020 and 2022 data (Section 4.2). An explicit accounting-reconciliation assertion in the pipeline (failing loudly

if a sector's computed output does not match the table's own `P1` row, in the spirit of the GVA check in Section 5.1) would be a reasonable next hardening step, though the reconciliation is already exact in the cases we checked.

The mobility term's conservation was fixed (Section 3.3): an earlier draft's $\bar{r}$ did not guarantee $\sum_i \mu(r_i - \bar{r})C_i = 0$; redefining $\bar{r}$ as the C_i-weighted mean of r_i makes this hold by construction, verified directly rather than assumed. *The rationing mechanism's demand term was also fixed*: an earlier draft applied A_{ij} directly to planned investment *expenditure* $\sigma_{c,j}S_j$, as if a unit of investment spending required the same input mix as a unit of physical output. Section 3.3's $\Delta x_j = \sigma_{c,j}S_j \cdot x_j/C_j$ resolves this by converting expenditure to an output-scale quantity via the sector's own capital-output ratio before A_{ij} is applied to it. One simplification remains, disclosed rather than resolved: the full reconstructed x_i is still treated as available for investment demand D_i before intermediate replacement (already counted in z_i) and consumption and other final demand are met — overstating investable supply, since no data source in the current pipeline provides sector-of-origin consumption baskets to net out. $D_i \leq x_i$ should therefore be read as an upper bound on investable supply, not a fully netted reproduction balance (Section 3.3).

The TSSI layer and the bifurcation analysis have a deliberately narrow scope. TSSI is implemented as a valuation of trajectories, not as a driver of them: the mobility term of Section 3.3 still uses simultaneist (Sraffian) profit rates, and a TSSI-driven variant of the dynamics is not implemented. The production lag τ is a single fixed constant, not a distribution of turnover times by sector, and depreciation d_j is valued at base prices rather than at historical acquisition prices. The bifurcation analysis concerns the two maps of Section 3.5; the accumulation ODEs themselves have no equilibria in levels and would need rewriting in normalized variables (capital shares, organic compositions) before their own bifurcations could be studied. Hessians (`system_hessian`) are available for the ODE right-hand side but not through the rationing fixed point, whose implicit-function derivative is first-order only.

The cross-dual extension and its cycles rest on strong, uncalibrated assumptions. Market prices adjust at a single speed β, final demand is a fixed basket (initial net-output shares) because the pipeline has no final demand by sector of origin, and neither β nor the variable-capital mobility μ_v is estimated. The Hopf points and cycles of Section 5.6 are therefore properties of the model class, not claims about any economy. The cycles approach the boundary of the simplex (sector shares of order 10^{-4}), which points to a missing braking mechanism such as entry of new capital or limits to exit. Numerically, two Hopf points within one continuation step escape detection (we use a step of at most 0.01 and check against a direct count of unstable eigenvalues), and bifurcations along the cycle branch depend on the step size, so we treat them as established only where direct simulation agrees. The chaos claim rests on finite-time Lyapunov exponents: they are robust to the initial condition, the integration tolerance and the run length in our checks, but they are not a computer-assisted proof.

Reproducibility is live-query reproducibility, not archival reproducibility. Every result in this paper is produced by re-running `generate_results.jl` against the live Eurostat and AMECO SDMX APIs (Availability, below), which is re-runnable but not immune to retroactive revisions in the underlying series, dataset vintages, or API structure — a live query today is not guaranteed to reproduce a number reported here indefinitely. A stronger reproducibility architecture would pair a tagged package release with cached raw API responses (or a frozen archival snapshot, e.g. on Zenodo) alongside the live-query path, letting a reader choose between reproducing this paper's exact reported numbers from frozen data and refreshing the analysis against current data; we have not built this yet.

7 Conclusion

MarxKapital demonstrates that a real, live, officially sourced data pipeline for Marxian value-theoretic calibration is practical using current European statistical infrastructure, and that an

exact per-sector quantity-rationing mechanism — solving a genuine fixed point, though not imposing hard market clearing at every instant (Section 3.3) — can be added to a reproduction-schema model without the numerical overhead of a formal differential-algebraic formulation. Applied to Portugal and Spain, the pipeline surfaces — and lets us diagnose and fix — two distinct measurement problems: a stock-flow conflation in the model's own output formula, caught by external review, and a resulting valuation artifact in the aggregate profit-rate series that we had originally overstated because of it. Corrected, the underlying trend is real but modest — a 7.9%–15.1% range for Portugal across three deflation conventions, not the near-fivefold swing an earlier, buggy version of this calibration reported. Two further defects surfaced by the same review process are also fixed and verified directly rather than merely asserted: the capital-mobility term now conserves aggregate constant capital exactly by construction, and the rationing mechanism's convergence and dispersion claims (Section 5.4) now rest on systematic sweeps of 270 and 50 random economies respectively, not a single hand-picked illustration. The exact Jacobian of the rationing map, computed by automatic differentiation, turned the convergence claim from a purely empirical one into a partly proved one: the Jacobian is triangular, its spectrum lies in $[-1, 0]$, damped iteration converges locally for every $\alpha \in (0, 1)$, and undamped iteration lies exactly on a flip boundary that a continuation analysis locates at the border-collision onset of rationing. A temporal single-system valuation layer with fixed production lag shows TSSI prices converging to simultaneist ones under frozen data in every economy tested. Rewriting the accumulation dynamics in normalized variables shows that the value-accounting model has no interior balanced growth; with market prices that respond to excess demand it does, and that balanced growth gives way to accumulation cycles through a supercritical Hopf bifurcation in the mobility of variable capital, at a threshold that rises with the wage share and with the speed of price adjustment; at higher mobility the cycles give way to coexisting attractors and to chaos. A fourth audit, of the technical matrix itself, found the most serious defect of the four: the pipeline had been using the wrong Eurostat dataflow, understating intermediate flows by more than an order of magnitude and producing an implausible spectral radius and maximum profit rate that this paper had, in an earlier version, reported without knowing why they looked wrong. Rebuilt from the correct table, the matrix reconciles exactly against its own published output figures and gives economically plausible values throughout (Section 4.2, Section 6.2). A fifth defect, in the dynamical model itself, was resolved rather than only disclosed: the coupling mechanisms' supply side omitted circulating constant capital entirely, and their investment-demand side applied the technical matrix to expenditure rather than output. Both are fixed by reconstructing genuine gross output from the model's own state at every solver evaluation (Section 3.1, Equation (2)) and converting planned investment expenditure to an output-scale quantity via each sector's capital-output ratio before applying A to it (Section 3.3) — neither requires new state or a heavier solver. We think this exercise is itself a useful illustration of what a live, multi-source statistical pipeline buys a computational Marxian model: not just convenience, but the ability to catch exactly this kind of measurement and modeling error before it is mistaken for an economic finding — provided the pipeline is checked as carefully as the theory it implements, and provided implausible intermediate results (a spectral radius an order of magnitude too low, in this case) are investigated rather than reported with a caveat and left for later. Extending the valuation-consistent comparison to more economies and archiving frozen data snapshots for stronger reproducibility guarantees (Section 6.2) are left for future work.

Availability

MarxKapital is released under the MIT license. The complete source code, including the empirical pipeline described here, is available at https://github.com/randsley/MarxKapital. Every figure and table in this paper is produced, without manual editing, by `paper/generate_results.jl`, which performs live queries against the Eurostat and AMECO SDMX APIs; re-running it re-

produces this paper's results from the same official data sources. The synthetic sweeps of Sections 5.4–5.5 are produced offline by `paper/monte_carlo.jl` and `paper/jacobian_bifurcation.jl`; the latter also writes every number quoted in Sections 3.4 and 5.5 to `paper/figures/jacobian_stats.tex`, which the paper reads directly.

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